



Physics-driven invertible architecture for unified forward-inverse modeling of magnetorheological dampers

Feng Zhang¹ · Minghui Zhu¹ · Zhiyuan Zou² · Jiahao Li³ · Benyuan Fu² · Honghui Zhang¹

Received: 25 May 2026 / Revised: 1 August 2026 / Accepted: 31 August 2026
© The Author(s), under exclusive licence to Springer Nature B.V. 2026

Abstract

High-fidelity forward force prediction and inverse current deduction are crucial for the semi-active vibration control of magnetorheological (MR) dampers. However, existing models suffer from bottlenecks like inadequate accuracy, intractable inverse solving, and a lack of physical interpretability. Therefore, this paper proposes a physics-driven invertible architecture (PDIA) for unified forward-inverse MR damper modeling. Unlike physics-informed neural networks relying on soft penalties, PDIA embeds physical mechanisms into the network topology as mandatory hard constraints. Specifically, complex damping dynamics are decoupled into a base response bias, a state modulation factor, and an electromagnetic mapping core, each subjected to targeted physical constraints. The core mapping is mathematically restricted to a global monotonic bijection, and a robust division operator is introduced to guarantee the absolute uniqueness and numerical stability of the inverse solution. Furthermore, a latent space alignment strategy formulates a joint loss within a differentiable domain, circumventing gradient truncation obstacles from discrete root-finding and enabling stable end-to-end joint optimization. Extensive experiments demonstrate PDIA's superior bi-directional performance. In zero-shot generalization tests, its forward and inverse MAE are as low as 0.0170 kN and 0.0211 A. Compared with other models, the PDIA exhibits significant advantages in accuracy, generalization, and stability. Moreover, hardware force-tracking experiments validate its precise tracking of dynamic target commands and practical feasibility. In summary, the proposed architecture mitigates the unconstrained limitations of purely black-box representations in non-linear modeling, offering a robust approach for the bidirectional modeling of magnetorheological dampers.

Keywords MR damper · Physical constraint · Invertible architecture · Unified forward-inverse modeling

1 Introduction

Magnetorheological (MR) dampers, characterized by fast response, low energy consumption, and continuously adjustable damping forces, serve as core actuators in semi-active vibration control systems [1–4]. In practical

applications, the desired damping force determined by upper-level control algorithms must be converted into physical excitation current commands for hardware execution. A highly reliable inverse model acts as the critical bridge facilitating this conversion [5, 6]. Concurrently, a high-fidelity forward model provides indispensable data

✉ Honghui Zhang
hhzhang@cqu.edu.cn

Feng Zhang
zfeng@cqu.edu.cn

Minghui Zhu
mhzhu@stu.cqu.edu.cn

Zhiyuan Zou
zyzou@cqu.edu.cn

Benyuan Fu
byfu@cqu.edu.cn

¹ Key Laboratory of Opto-Electronic Technology and Systems of the Ministry of Education, Chongqing University, Chongqing, China

² College of Mechanical Engineering, Chongqing University of Technology, Chongqing, China

³ CAS Key Laboratory of Mechanical Behavior and Design of Materials, Department of Modern Mechanics, University of Science and Technology of China (USTC), Hefei 230027, Anhui, China

support for virtual simulation and state observation [7, 8]. Therefore, establishing highly accurate forward and inverse dynamic models is a fundamental prerequisite for realizing closed-loop control of MR dampers [9, 10]. However, governed by complex internal microscopic rheological mechanisms, the macroscopic mechanical responses of MR dampers exhibit pronounced strong nonlinearity and severe hysteresis [11, 12], posing formidable challenges to accurate modeling.

To address these challenges, traditional MR damper modeling has primarily relied on parameterized phenomenological models. These methods conceptualize the damper as a combination of macroscopic mechanical elements, derive parameterized equations, and identify core parameters via optimization algorithms [13]. Representative works include the Bingham model [14, 15], Bouc-Wen model [16–18], hyperbolic tangent model [19, 20], polynomial model [21], RC operator model [22–24], and their various extensions [25–27]. However, such models suffer from an inherent structural dilemma: a trade-off between representation accuracy and computational complexity [28]. Consequently, they struggle to precisely capture high-order nonlinear hysteretic features under complex operating conditions [29]. More critically, constrained by their forward-modeling paradigm and complex coupled differential equations [30], deriving explicit analytical inverses for the control current from these phenomenological models is mathematically intractable.

To bypass complex mechanistic derivations and capture the nonlinear hysteresis of MR dampers, researchers have proposed various non-parametric neural network models [31–34], essentially treating the damper as a black box. Representative architectures include FNN [35], RNN [36], NLSTM [37], and various customized networks—such as the block-structured Hammerstein model [5], the optimized MEA-BP network [32], and the lag-compensated TF-BPNN model [38]. Leveraging their universal approximation capabilities, these models can accurately predict nonlinear details. However, purely data-driven models inherently rely on implicit probabilistic memory and lack fundamental physical meaning [39]. Consequently, they are prone to generating physically unreasonable predictions under unseen conditions. Furthermore, lacking explicit structural invertibility, black-box models necessitate independent bi-directional modeling. This separated approach not only increases computational overhead but also easily induces mapping conflicts due to independently updated network weights.

To enhance model interpretability, Physics-Informed Neural Networks (PINNs) have been widely introduced into nonlinear engineering predictions [40–42]. PINNs guide models to learn physics-compliant mappings by incorporating residual terms from governing differential

equations into the loss function [43, 44]. However, current PINNs predominantly rely on soft constraint penalties. This indirect intervention, dependent on gradient optimization trade-offs, inherently lacks mandatory physical constraints within the underlying network topology.

To overcome the aforementioned bottlenecks, this paper presents a physics-driven invertible architecture (PDIA) for the unified forward-inverse modeling of MR dampers. Discarding the conventional paradigm of treating physical laws as soft loss penalties, the PDIA innovatively embeds macroscopic physical mechanisms deep into the network topology as mandatory hard constraints. Leveraging variable-separation characteristics, the complex dynamic equation is rigorously decoupled into a mechanically dominated base-response bias and a state-modulation factor, and a transient-current-dominated monotonic electromagnetic-mapping core. Each component is subjected to targeted physical constraints. Through this physically isomorphic mapping, unified forward-inverse modeling is achieved with both high accuracy and stability within a single network framework. The main contributions are as follows:

- (1) Proposing a physics-forced hard-constrained architecture that overcomes the limitations of black-box modeling. Macroscopic physical laws are translated into hard topological constraints on the underlying latent variables of the network. This preserves the superior nonlinear approximation capabilities of deep learning while endowing the model with extremely high physical interpretability.
- (2) Introducing a monotonic bijective constraint and a robust division operator to endow the model with globally invertible attributes. A strict monotonic bijective constraint is imposed on the electromagnetic core gain to construct an invertible space. Simultaneously, a robust division operator is integrated to facilitate accurate current inversion across the entire velocity range, thereby endowing the model with globally reliable and unique invertibility.
- (3) Establishing a unified forward-inverse optimization based on latent space alignment. Operating within a continuously differentiable high-dimensional latent domain, the Huber loss is utilized to unblock the global gradient backpropagation pathway. This strategy guarantees the absolute mathematical equivalence of the forward and inverse mappings and endows the model with dynamic robustness under extreme unseen zero-shot conditions.

The remainder of this paper is organized as follows: Sect. 2 derives the invertible expressions and physical constraints. Section 3 details the overall PDIA architecture and the unified training strategy. Section 4

comprehensively validates the performance of the proposed framework through multi-condition experiments. Finally, Sect. 5 concludes the paper.

2 Theoretical analysis of the invertible model

The fundamental objective of the PDIA is to reformulate the damping force into a physically meaningful, structured expression based on underlying mechanisms. By examining the rheological behavior of MR fluids, electromagnetic responses, and energy dissipation characteristics, this section analyzes the fundamental physical constraints governing the MR damper dynamics and subsequently formulates the invertible dynamic expression.

2.1 Basic physical constraint analysis

A. Rheological Mechanisms of MR Fluids: The constitutive behavior of MR fluids is typically idealized using the Bingham plastic model:

$$\tau = \tau_y(B) \operatorname{sgn}(\dot{\gamma}) + \eta \dot{\gamma} \quad (1)$$

where τ denotes the total shear stress, $\tau_y(B)$ is the dynamic yield stress modulated by the magnetic induction intensity B , η is the plastic dynamic viscosity of the fluid, $\dot{\gamma}$ is the shear strain rate, and $\operatorname{sgn}(\cdot)$ is the sign function.

At the microphysical level, the structural shear strength of the magnetic particle chains is positively correlated with the applied magnetic field intensity. Thus, the fluid yield stress response to the magnetic field shows a strictly monotonic increase across the entire domain. This leads to the first physical constraint:

Physical constraint I (Intrinsic Magnetization Monotonicity): The partial derivative of the yield stress with respect to the magnetic induction intensity must be strictly non-negative:

$$\frac{\partial \tau_y}{\partial B} \geq 0 \quad (2)$$

That is, an increase in the magnetic field will never cause a decrease in the fluid yield stress.

B. Electromagnetic Response: The internal magnetic field of the MR damper is induced by the control current I in the excitation coil. In the low-current region, the electromagnetic response $B(I)$ increases with the current I . However, constrained by the physical limits of the magnetic core and fluid particles, the system gradually enters magnetic saturation when the current exceeds a certain threshold. This establishes the second physical constraint:

Physical constraint II (Zero Baseline and Saturation Bound): Governed by the causality of electromagnetic induction, the induced magnetic field vanishes when the control current is removed (neglecting remanence):

$$B(0) = 0 \quad (3)$$

Furthermore, due to the magnetic saturation effect, the second derivative of the electromagnetic response with respect to the control current I is subject to an asymptotic bound, indicating a monotonically decreasing growth rate.

C. Dissipation Characteristics: The primary function of an MR damper is the dissipation of mechanical energy. Under standard testbed conventions, the output damping force F and the piston velocity v share the same directional sign convention. Therefore, the dissipated mechanical power must constantly remain non-negative. This establishes the third physical constraint:

Physical constraint III (Energy Dissipation): The mechanical work executed by the output force of the MR damper must strictly manifest as energy dissipation. Under any dynamic state, the system must satisfy the following inequality:

$$P = F \cdot v \geq 0 \quad (4)$$

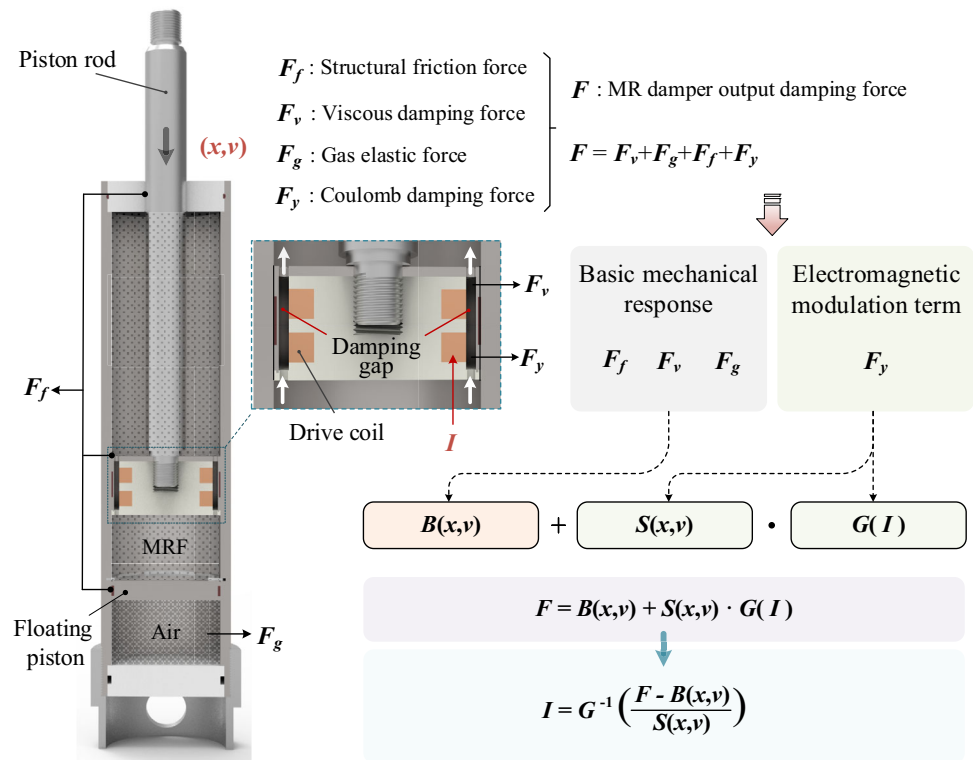
2.2 Dynamic expression of the invertible structure

As shown in Fig. 1, an analysis of the macroscopic non-linear dynamics reveals that the damping force of MR dampers originates from two fundamentally distinct physical mechanisms: a base mechanical response driven by piston motion, and an additional Coulomb damping force modulated by the magnetic field. Consequently, the output force is formulated as a coupling of mechanical and electromagnetic terms. Considering the structural friction and inherent hysteresis present in practical MR dampers, the macroscopic output force F is physically abstracted as a coupled nonlinear function governed jointly by mechanical (displacement x , velocity v) and electromagnetic (current I) excitations:

$$F(x, v, I) = F_m(x, v) + F_e(x, v, I) \quad (5)$$

where $F_m(x, v)$ denotes the base dynamic term associated with mechanical motion, and $F_e(x, v, I)$ represents the additional damping term induced by electromagnetic excitation.

Further analysis reveals that the electromagnetic modulation fundamentally alters the yield stress of the MR fluid, thereby scaling the amplitude of the Coulomb damping force. In contrast, the acting direction is strictly dictated by the mechanical motion state. Based on this variable separation characteristic, the electromagnetic

Fig. 1 Invertible kinetic structure of MR damper

modulation term is rigorously decoupled into the product of a “motion state modulation factor” and a “pure electromagnetic response function”:

$$F_e(x, v, I) = S(x, v)G(I) \quad (6)$$

where $S(x, v)$ denotes the state modulation function, characterizing the modulating effect of the motion state on the magnetization response, and $G(I)$ represents the electromagnetic mapping function, describing the nonlinear magnetization characteristics induced by the current excitation.

Consequently, the output force of the MR damper is reconstructed into the following structured expression:

$$F(x, v, I) = B(x, v) + S(x, v)G(I) \quad (7)$$

where $B(x, v)$ is the base dynamic function, describing the fundamental mechanical response under zero-current conditions. Notably, due to the pronounced dynamic evolution during the formation and breakage of micro-particle chains within the MR fluid, the functions $B(\cdot)$ and $S(\cdot)$ are not simple instantaneous functions in the actual system. Instead, they implicitly depend on the historical motion trajectory of the system and the magnetization process.

To ensure the physical rigor of this proposed architecture, the three aforementioned fundamental physical constraints are mathematically translated into the following topological rules:

a. Magnetization monotonicity constraint: The electromagnetic mapping function must remain strictly monotonically increasing across the entire domain, yielding:

$$\frac{dG(I)}{dI} > 0 \quad (8)$$

b. Zero-baseline and saturation bound: The magnetic modulation effect vanishes entirely when the control current is zero. Furthermore, as the current continuously increases, the system gradually transitions into magnetic saturation, indicating:

$$G(0) = 0, \frac{d^2G(I)}{dI^2} \leq 0 \quad (9)$$

c. Energy dissipation constraint: Under any kinematic state, the work done by the additional Coulomb damping force must strictly manifest as energy dissipation, satisfying:

$$S(x, v) \cdot v \geq 0 \quad (10)$$

Since the electromagnetic mapping function $G(I)$ has a monotonicity constraint imposed on it, the current can be inversely derived from the damping force expression in the following form:

$$I = G^{-1}\left(\frac{F - B(x, v)}{S(x, v)}\right) \quad (11)$$

Therefore, this model mathematically constitutes an invertible dynamic representation, enabling the establishment of a stable, unique one-to-one mapping relationship between the damping force and the control current.

3 Physics-driven invertible architecture

Using the established invertible dynamic expression as the physical skeleton, this section constructs the physics-driven invertible architecture (PDIA). This architecture achieves integrated modeling for high-fidelity forward dynamic prediction and inverse current deduction within a unified network framework.

3.1 Overall structure

Considering that the transient output damping force of MR dampers is governed by the system's historical evolution trajectory, to transform the structured expression described in Eq. (7) into a physics-driven invertible network, it is sequentially extended in the discrete time dimension. The resulting neural network mapping equation is formulated as follows:

$$F_t = \text{Bias}(\mathbf{X}_{t-L:t}, \Psi_{t-L:t}) + \text{Shape}(\mathbf{X}_{t-L:t}, \Psi_{t-L:t}) \cdot G(I_t) \quad (12)$$

where $\mathbf{X}_{t-L:t} = [x, v]_{t-L:t}^T$ represents the historical mechanical kinematic sequence of length L , and $\Psi_{t-L:t}$ denotes the auxiliary historical input sequence of length L used to assist network inference. For integrated forward-inverse modeling and to fully separate bi-directional information flows: in forward damping force prediction, $\Psi_{t-L:t}$ is the input historical current sequence, whereas in inverse current deduction, $\Psi_{t-L:t}$ is the input historical target force sequence. $\mathbf{X}_{t-L:t}$ and $\Psi_{t-L:t}$ are used together to precisely extract the damper's nonlinear delayed features. Furthermore, the entire mapping equation is strictly constrained by the three physical constraints mentioned above.

The PDIA is designed using Eq. (12) as its foundational topological criterion, with the specific structure illustrated in Fig. 2. The overall network structure is primarily divided into a state observer and an invertible electromagnetic core. Rather than enforcing a global, static bijective mapping across the entire dynamic state-space (which would non-physically smooth out historical asymmetries), the PDIA adopts a decoupled architecture that separates dynamic non-linearities from mathematical invertibility.

The state observer is mainly utilized to extract the base dynamic term *Bias* and the motion state modulation factor *Shape*. Specifically, this observer is designed to fully accommodate the complex, history-dependent, and

asymmetrical dynamic behaviors—such as microstructural magnetic chain rupture, fluid hysteresis, and mechanical energy dissipation. By utilizing historical kinematics, the observer models these memory effects without imposing any static bijective constraints, thereby preserving the raw physical sovereignty of the dynamic path-dependency.

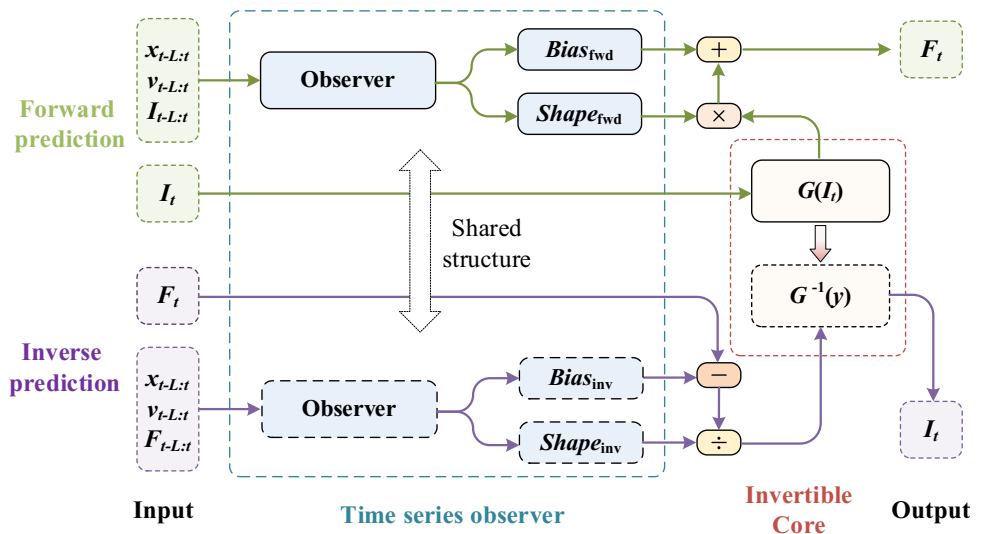
Conversely, the invertible electromagnetic core focuses on describing the nonlinear characteristics between the transient current and the magnetization response. To ensure the uniqueness of the inverse deduction, strict mathematical monotonic hard constraints are imposed on this module. This is physically consistent because, as the control current increases, the magnetic induction strength and the resulting yield stress of the MR fluid monotonically increase. Crucially, the three fundamental physical constraints proposed earlier serve as explicit topological restrictions, which are directionally injected into the functional modules through structural definitions. This highly unified design of physics and structure ensures that the model maintains physical self-consistency under any input sample.

Based on the physical architecture's invertibility, the designed overall network enables unified forward-inverse modeling. In the forward force prediction path, the kinematic and current sequences are fed into the forward observer, extracting Bias_{fwd} and $\text{Shape}_{\text{fwd}}$ with hysteretic memory. Simultaneously, the transient current I_t is passed through the invertible electromagnetic core $G(\cdot)$ to obtain the output $G(I_t)$. Ultimately, these three physical features are processed via multipliers and adders at the network terminal to forward-reconstruct the predicted damping force F_t at the current time step. In the inverse current deduction path, since the current is unknown, current information is strictly masked at the input end. Instead, the kinematic and target force sequences are entered into the inverse observer to solve for the inverse-specific Bias_{inv} and $\text{Shape}_{\text{inv}}$. Subsequently, subtractors and dividers are utilized according to the inverse formula to calculate the target latent variable y . Finally, the core operator's inverse mapping $G^{-1}(y)$ gives the transient current I_t needed to meet the target force.

3.2 State observer

Based on the network mapping established in Eq. (12), the specific structure of the forward and inverse state observers designed for the PDIA is illustrated in Fig. 3. Its backbone feature extraction module consists of a two-layer Gated Recurrent Unit (GRU). To guarantee the physical logic of the bi-directional inference, the forward and inverse observers adopt a structural reuse mechanism to map the unified internal hysteretic laws of the damper. However, regarding network parameters, they maintain independent weights to completely isolate information leakage and

Fig. 2 Schematic diagram of the overall network structure of the PDIA



input domain confusion during the bi-directional inference process. The inputs to the forward observer are $x_{t-L:t}$, $v_{t-L:t}$, and $I_{t-L:t}$, with outputs of $Bias_{fwd,t}$ and $Shape_{fwd,t}$; whereas the inputs of the inverse observer are $x_{t-L:t}$, $v_{t-L:t}$, and $F_{t-L:t}$, with outputs of $Bias_{inv,t}$ and $Shape_{inv,t}$. This asymmetric input–output configuration defines the core distinction between the two observers while maintaining structural similarity. Both observers share the same internal recurrent network architecture to extract the macroscopic state parameters representing the history-dependent hysteresis of the MR damper. However, the forward observer

utilizes the control current sequence $I_{t-L:t}$ as an active input to forward-reconstruct the damping force, whereas the inverse observer strictly masks the current input. Instead, the inverse observer ingests the required target force sequence $F_{t-L:t}$ along with kinematics to solve for the inverse-specific $Bias_{inv,t}$ and $Shape_{inv,t}$. This asymmetric decoupling prevents mathematical circular dependencies during current deduction, ensuring that the subsequent algebraic inversion remains well-conditioned and numerically stable during real-time executions. Relying on the input time series, the observer utilizes the two-layer GRU

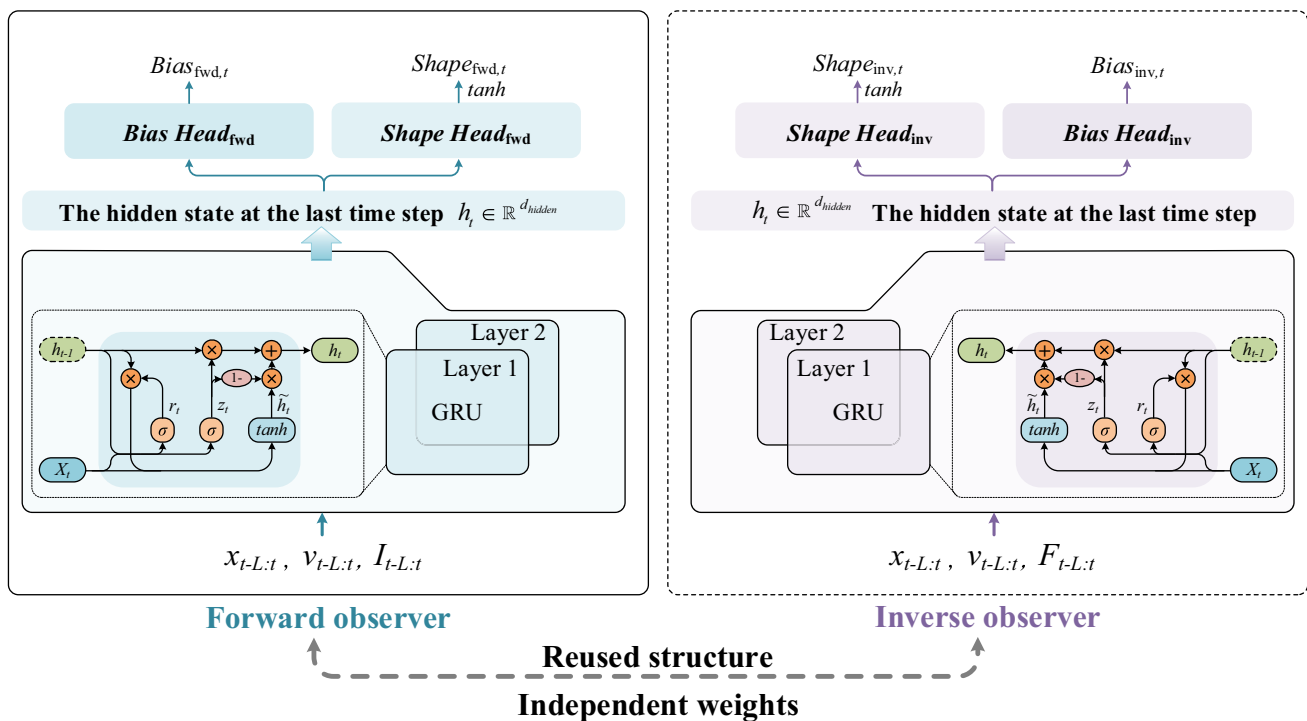


Fig. 3 Schematic diagram of the structure of the state observer

architecture in the discrete time dimension, where the first GRU layer focuses on capturing fundamental motion features, and the second GRU layer further abstracts higher-order dynamic states.

Within a time sliding window of length L , for the time step $\tau \in [t-L, t]$, the input feature vector of the observer is defined as $\mathbf{V}_\tau = [x_\tau, v_\tau, \psi_\tau]^T \in \mathbb{R}^3$. Here, the auxiliary variable ψ_τ is directionally assigned based on the inference flow (i.e., $\psi_\tau = I_\tau$ in the forward observer, and $\psi_\tau = F_\tau$ in the inverse observer). The recurrent updating process within the GRU is expressed as:

$$\begin{cases} \mathbf{r}_\tau = \sigma(\mathbf{W}_r \mathbf{V}_\tau + \mathbf{U}_r \mathbf{h}_{\tau-1} + \mathbf{b}_r), \mathbf{z}_\tau = \sigma(\mathbf{W}_z \mathbf{V}_\tau + \mathbf{U}_z \mathbf{h}_{\tau-1} + \mathbf{b}_z) \\ \tilde{\mathbf{h}}_\tau = \tanh(\mathbf{W}_h \mathbf{V}_\tau + \tilde{\mathbf{U}}_h (\mathbf{r}_\tau \odot \mathbf{h}_{\tau-1}) + \mathbf{b}_h), \mathbf{h}_\tau = (1 - \mathbf{z}_\tau) \odot \mathbf{h}_{\tau-1} + \mathbf{z}_\tau \odot \tilde{\mathbf{h}}_\tau \end{cases} \quad (13)$$

where σ denotes the Sigmoid activation function, $\mathbf{r}_\tau$ and $\mathbf{z}_\tau$ represent the reset gate and update gate, respectively, and $\mathbf{h}_{\tau-1}$ is the hidden state memory from the previous time step. $\mathbf{W}_r, \mathbf{W}_z, \mathbf{W}_h$ and $\mathbf{U}_r, \mathbf{U}_z, \mathbf{U}_h$ are the learnable connection weight matrices, while $\mathbf{b}_r, \mathbf{b}_z, \mathbf{b}_h$ are the corresponding bias vectors.

As illustrated in Fig. 3, after extracting the hidden state that contains hysteretic characteristics, the network generates outputs using two parallel linear readout heads. To ensure the network strictly adheres to macroscopic mechanical laws, the design of these readout heads is subjected to mandatory physical constraints.

$$\text{Bias}_t = \mathbf{W}_{\text{bias}} \mathbf{h}_t + \mathbf{b}_{\text{bias}} \quad (14)$$

$$\text{Shape}_t = \tanh(\mathbf{W}_{\text{shape}} \mathbf{h}_t + \mathbf{b}_{\text{shape}}) \quad (15)$$

The authentic zero-field base damping force comprises not only the fluid viscous damping but also the elastic restoring force generated by the compensation accumulator. Crucially, this elastic force can perform positive work on the system during the piston rebound stroke. Therefore, the Bias branch in Eq. (14) explicitly excludes any activation function, granting the network the necessary degrees of freedom required to reconstruct complex base dynamic characteristics.

According to the previously established “energy dissipation” physical constraint, the additional Coulomb damping force must continuously perform negative work to dissipate mechanical energy. Since the subsequent electromagnetic core $G(I_t)$ is strictly constrained as a non-negative mapping ($G > 0$), the modulation factor must constantly satisfy the sign condition $\text{Shape}_t \cdot v_t \geq 0$. To accommodate the work characteristics during alternating compression and rebound strokes (where v_t changes sign), the *Shape* readout head in Eq. 15 is strictly equipped with a symmetric tanh activation function bounded within $[-1, 1]$. This activation function has two functions. First, it

structurally restricts the amplification ratio of the additional damping force to prevent non-physical divergent explosions. Second, it endows the operator with the ability to output positive and negative values across the zero-crossing point. As a result, the network can autonomously learn the physical reversal laws of the damping force direction. It can smoothly transition through the velocity zero-crossing dead-zone. Thus, at the underlying structural level, it fundamentally eliminates the generation of non-physical active thrusts.

3.3 Structure of the invertible core

The invertible core serves as the key part of the PDIA for achieving unified forward-inverse modeling. Based on electromagnetic laws, to enable the core to accurately capture the current’s nonlinear saturation and provide a singularity-free global analytical inverse, it is designed with mandatory monotonic constraints. Its specific topology and internal logic are illustrated in Fig. 4.

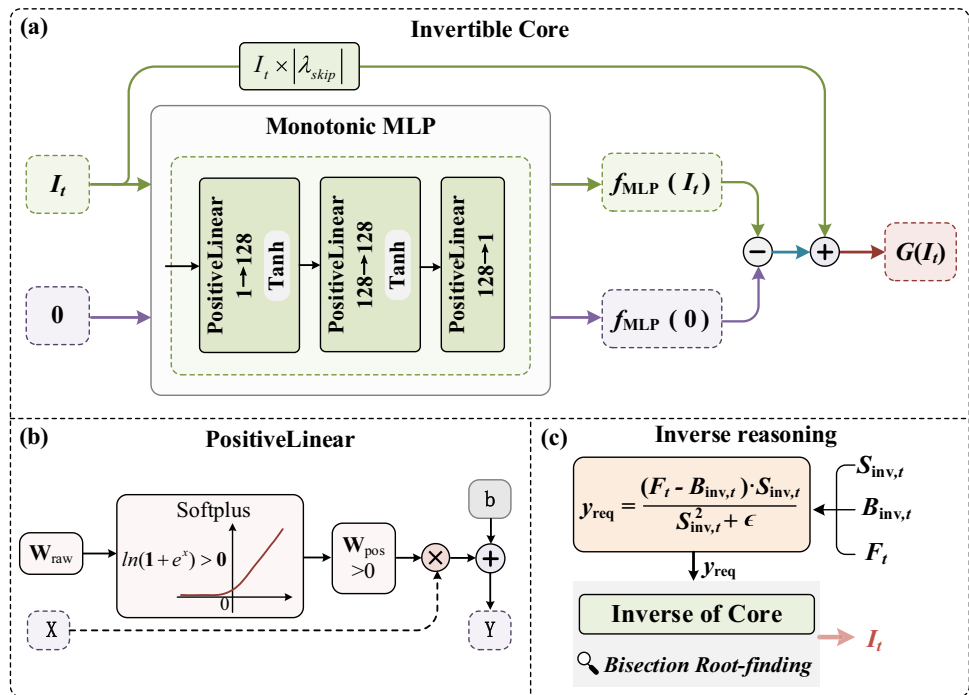
As shown in Fig. 4a, the invertible core primarily adopts a monotonic multilayer perceptron (MLP) architecture integrating PositiveLinear layers and a Skip-connection. The input transient current I_t and the zero-baseline constant 0 are fed into the MLP in parallel. Their respective outputs are algebraically subtracted to eliminate background zero-drift. Simultaneously, coupled with a linear skip-connection, the final electromagnetic response $G(I_t)$ is yielded. Herein, the PositiveLinear layer is a customized fully connected layer. Distinct from traditional fully connected layers, all raw network weights $\mathbf{W}_{\text{raw}}$ are strictly mapped to the positive space via a Softplus activation mapping before executing the affine transformation, as detailed in Fig. 4b.

As shown in Fig. 4c, during the inverse deduction of the current, the target latent variable y_{req} is initially computed utilizing a robust division operator combined with a stabilizing factor ε , thereby avoiding the zero-division singularity within the physical dead-zone. Subsequently, the exact control current I_t is analytically deduced within the monotonic space employing the bisection root-finding algorithm.

The specific design principles, monotonicity proof, and physical significance of this invertible core are detailed below.

3.3.1 Design of forward mapping

To satisfy the physical constraint of the “zero baseline and saturation bound”, the core network $G(I)$ is structurally tailored. First, a zero-deduction mechanism is introduced to address the zero baseline $G(0) = 0$. The network takes both the transient current I_t and a zero-baseline constant 0

Fig. 4 Schematic diagram of the invertible core architecture

as parallel inputs. A structured subtraction $f_{\text{MLP}}(I_t) - f_{\text{MLP}}(0)$, is then applied. This operation completely neutralizes the zero-input response caused by hidden layer biases. Consequently, the zero baseline is structurally enforced.

Second, to characterize the saturation bound at high currents $d^2G(I)/dI^2 \leq 0$, the core network employs the tanh activation function. This function features a strictly negative second derivative ($\sigma'' < 0$). As the input current increases, its first-order gradient progressively decays. This macroscopically replicates the nonlinear asymptotic saturation of the fluid's yield stress. However, absolute physical saturation (i.e., the local gradient dropping to zero) would inevitably cause multi-valued mappings. This would destroy the uniqueness of the inverse root-finding. To prevent this, a linear skip connection ($I_t \times |\lambda_{\text{skip}}|$) as illustrated in Fig. 4a is introduced as a gradient-bounding mechanism. Ultimately, the electromagnetic core function is formulated as:

$$G(I_t) = [f_{\text{MLP}}(I_t) - f_{\text{MLP}}(0)] + I_t \cdot |\lambda_{\text{skip}}| \quad (16)$$

where $G(I_t)$ denotes the electromagnetic core mapping function at time step, and I_t is the input control current. $f_{\text{MLP}}(\cdot)$ represents the output function of the monotonic Multi-Layer Perceptron (MLP). $|\lambda_{\text{skip}}|$ is the non-negative skip-connection weight. This design allows the model to maintain a strictly positive, albeit marginal, gradient lower bound in the high-current saturation region. Therefore, it ensures global strict monotonicity while perfectly preserving the physical saturation characteristics.

3.3.2 Global monotonic bijection

To enforce the physical constraint of “intrinsic magnetization monotonicity $dG(I)/dI > 0$ ” proposed earlier, this architecture redesigns the weight mapping mechanism of traditional fully connected layers. During the forward propagation of the PositiveLinear operator, all learnable raw weight matrices $\mathbf{W}_{\text{raw}}$ are strictly mapped to strictly positive values via the Softplus function:

$$\mathbf{W}_{\text{pos}} = \ln(1 + \exp(\mathbf{W}_{\text{raw}})) > 0 \quad (17)$$

where $\mathbf{W}_{\text{raw}}$ and $\mathbf{W}_{\text{pos}}$ represent the raw weight matrix and the strictly positive mapped weight matrix of the network layers. Assuming the MLP contains L layers, with the input layer denoted as $h_0 = I \in \mathbb{R}^1$, and the hidden layers indexed by $l = 1, 2, \dots, L-1$. The sequential forward propagation can be defined as:

$$z_l = \mathbf{W}_{\text{pos},l} h_{l-1} + b_l, h_l = \tanh(z_l) \in \mathbb{R}^{d_l} \quad (18)$$

where z_l represents the pre-activation state vector of the l -th layer, and b_l is the corresponding bias vector. h_l denote the activated hidden state vectors of the l -th layer. $\mathbf{W}_{\text{pos},l}$ is the strictly positive weight matrices of the l -th hidden layer. Non-linear Activation Output Layer:

$$f_{\text{MLP}}(I) = h_L = \mathbf{W}_{\text{pos},L} h_{L-1} + b_L \in \mathbb{R}^1 \quad (19)$$

According to the chain rule of matrix calculus, the global first-order derivative of the network output $f_{\text{MLP}}(I)$ with respect to the input I can be expanded as the continuous product of the Jacobian matrices of each layer:

$$\frac{df_{\text{MLP}}(I)}{dI} = \frac{dh_L}{dh_{L-1}} \cdot \frac{dh_{L-1}}{dh_{L-2}} \cdots \frac{dh_1}{dh_0} \quad (20)$$

Substituting the specific derivative expressions of each layer yields:

$$\frac{df_{\text{MLP}}(I)}{dI} = \mathbf{W}_{\text{pos},L} \cdot \left[\prod_{l=1}^{L-1} \text{diag}(\sigma'(z_l)) \cdot \mathbf{W}_{\text{pos},l} \right] \quad (21)$$

where $\text{diag}(\cdot)$ is a diagonal matrix with the vector elements on its main diagonal, and $\sigma'(z) = 1 - \tanh^2(z)$ represents the derivative of the hidden layer activation function.

From Eq. (17), every element within the weight matrices is strictly greater than 0. Simultaneously, since the $\tanh(z) \in (-1, 1)$, its derivative $\sigma'(z) \in (0, 1]$ remains strictly positive across the real domain. Consequently, the diagonal elements of the matrix $\text{diag}(\sigma'(z_l))$ are strictly positive. Under the rules of matrix multiplication, the continuous product of strictly positive matrices and positive diagonal matrices inevitably results in a matrix where all elements are strictly greater than 0. That is:

$$\frac{df_{\text{MLP}}(I)}{dI} > 0 \quad (22)$$

Furthermore, taking the derivative of the complete core operator $G(I)$ yields:

$$\frac{dG(I)}{dI} = \frac{df_{\text{MLP}}(I)}{dI} - 0 + |\lambda_{\text{skip}}| > |\lambda_{\text{skip}}| \geq 0 \quad (23)$$

Because the network derivative possesses a strictly positive lower bound $|\lambda_{\text{skip}}|$ across the entire real domain, this neural network mapping constitutes a strict global monotonic bijective space. This mathematically guarantees a unique solution for the inverse current deduction.

3.3.3 Inverse inference calculation

Having established the global monotonic bijection of the invertible core, the inverse current deduction is successfully reduced to a one-dimensional (1D) analytical root-finding problem. According to the network mapping equation, the target electromagnetic latent variable can be algebraically expressed as $G(I_t) = (F_t - \text{Bias}_{\text{inv}})/\text{Shape}_{\text{inv}}$. However, in real-world physical inference scenarios, when the damper operates near the piston velocity zero-crossing points (i.e., physical dead-zones), the state modulation factor approaches zero $\text{Shape}_{\text{inv}} \rightarrow 0$. Consequently, any minuscule prediction noise in the numerator is infinitely amplified by the vanishing denominator, triggering catastrophic computational divergence.

To completely overcome this issue, direct division is discarded during inverse inference. Instead, a robust division operator featuring a minuscule stabilizing factor ε (set

to 10^{-4} in this study) is introduced to safely compute the target electromagnetic latent variable y_{req} :

$$y_{\text{req}} = \frac{(F_t - \text{Bias}_{\text{inv}}) \cdot \text{Shape}_{\text{inv}}}{\text{Shape}^2 + \varepsilon} \quad (24)$$

When operating far from the dead-zones, this operator is asymptotically equivalent to the exact algebraic division. Conversely, as the system approaches the dead-zones, the factor ε in the denominator effectively caps the output upper bound, smoothly suppressing the zero-division singularity. Once the target latent variable y_{req} is obtained, its inverse mapping $G^{-1}(\cdot)$ inevitably exists and is unique, given that $G(I)$ is strictly monotonic and continuous. By using the bisection root-finding algorithm, which has very low computational complexity, the system converges absolutely within a predefined tolerance. This explicitly determines the unique and exact inverse control current I_t .

From a non-linear dynamics perspective, the role of the stabilizing factor ε extends far beyond a mere computational patch; it represents a profound physical regularization mechanism. In the near-zero-velocity region, the damper's resistance is primarily dominated by the inherent mechanical friction of the cylinder seals (exhibiting stick-slip phenomena), whereas the microstructural yield transition of the MR fluid, although present, exerts a relatively weak and minor influence under such near-zero shear rates. Modeling this transition with an exact algebraic inverse (i.e., $\varepsilon=0$) erroneously yields a discontinuous step function, implying an infinite transient stiffness.

By contrast, the introduction of ε mathematically transforms this abrupt transition into a physically plausible micro-velocity boundary layer (i.e., a micro-slip region). When the system operates within this threshold ($|v| \sim \varepsilon$), the behavior of ε is akin to a micro-viscoelastic spring. It qualitatively represents the pre-sliding elastic deformation of the contacting seal asperities—which constitutes the primary source of resistance before macroscopic sliding occurs—while also incorporating the minor shear deformation exhibited by the unruptured magnetic particle chains. By embedding this physical regularization mechanism, the model successfully preserves the system's authentic intrinsic expression under threshold conditions. Simultaneously, it effectively caps the output upper bound, thereby smoothly suppressing the zero-division singularity.

3.4 Unified forward-inverse modeling strategy

The ultimate challenge for unified forward-inverse modeling is to realize an end-to-end training strategy that integrates forward force prediction and inverse current deduction. The integrated computational graph and error

backpropagation mechanism designed in this paper are illustrated in Fig. 5. During global forward computation, two physical pathways are activated. One is that the forward observer Obs_{fwd} works with the electromagnetic core to reconstruct the predicted damping force. The other is that the inverse observer Obs_{inv} calculates the target electromagnetic latent variable with the robust division operator. These two pathways share identical parameters of the unique invertible core, achieving a strict structural unification of the forward and inverse mappings.

Within the well-defined monotonic range of $G(\cdot)$, the bisection method demonstrates extremely high efficiency and absolute reliability during the inference phase. However, during the backpropagation training of the model, the underlying discrete numerical search logic of the bisection method blocks the chain rule of gradients in the loss function. This gradient truncation effect prevents the error from propagating back to the inverse observer, causing its weights to be deadlocked during the training process.

To strictly this gradient truncation problem, this paper proposes a latent space alignment training strategy. As shown in Fig. 5, instead of computing errors in the non-differentiable current space, this strategy inputs the actual current from the dataset into the shared-weight monotonic core. This maps out an absolutely true and differentiable electromagnetic latent feature baseline. The algorithm then directly measures the alignment between this baseline and the target latent feature deduced by the inverse observer. Small high-frequency numerical noises from the division

operator exist in physical dead zones. So, the inverse alignment loss abandons the overly outlier-sensitive Mean Squared Error (MSE) and adopts the Huber Loss (Smooth L1 Loss) that combines robustness with local smoothness:

$$\mathcal{L}_{\text{inv}} = \frac{1}{N} \sum_{t=1}^N \text{Huber}_{\beta}(y_{\text{pred},t}, y_{\text{true},t}) \quad (25)$$

where $\mathcal{L}_{\text{inv}}$ denote the inverse alignment loss, N represents the total number of training samples within a mini-batch, and t is the sample index. Huber_{β} represents the Huber loss operator. Conversely, the loss for the forward dynamic reconstruction retains the standard MSE :

$$\mathcal{L}_{\text{fwd}} = \frac{1}{N} \sum_{t=1}^N \|F_{\text{pred},t} - F_{\text{true},t}\|_2^2 \quad (26)$$

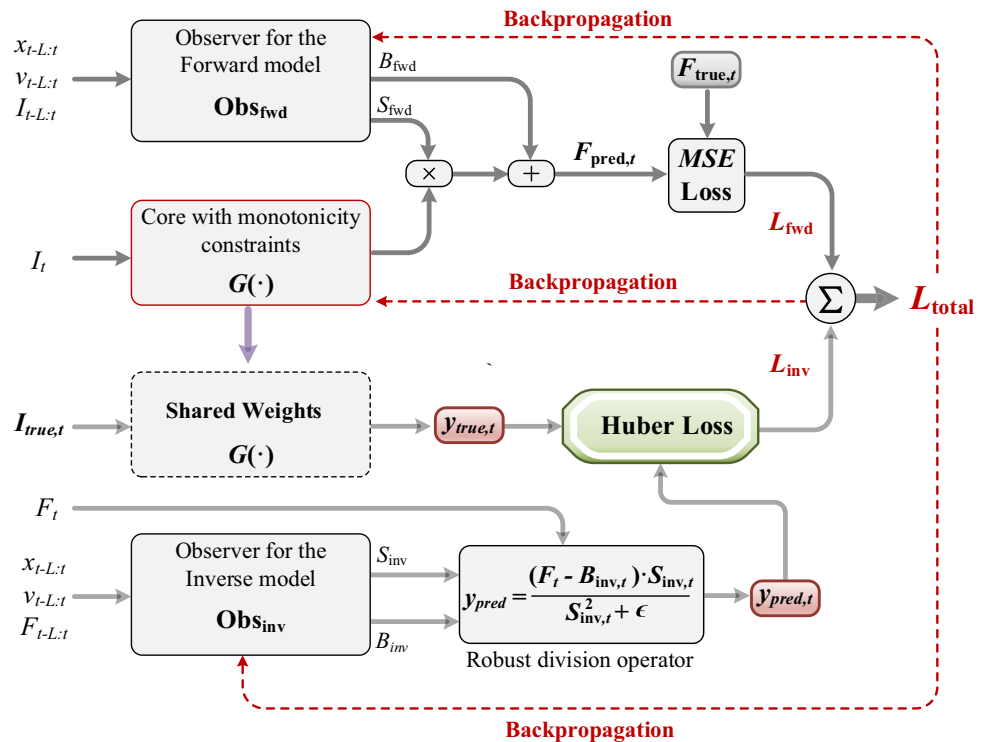
where $\mathcal{L}_{\text{fwd}}$ denote the forward dynamic reconstruction loss. Ultimately, the global joint loss function of the system is defined as the weighted sum of the forward and inverse errors:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{fwd}} \mathcal{L}_{\text{fwd}} + \lambda_{\text{inv}} \mathcal{L}_{\text{inv}} \quad (27)$$

where $\mathcal{L}_{\text{total}}$ denote the global joint loss function of the bidirectional system, λ_{fwd} and λ_{inv} are the trade-off coefficients for the bi-directional gradients.

As indicated by the red dashed backpropagation pathways in Fig. 5, this joint loss function design effectively converts the non-differentiable, complex inverse problem into an equivalent feature-matching optimization problem

Fig. 5 Unified forward-inverse modeling training strategy



within a differentiable latent domain. This approach ensures strict mathematical equivalence between forward and inverse mappings. It also bypasses the gradient truncation from the bisection method, achieving unobstructed full-pathway gradient backpropagation. Thus, it guarantees stable end-to-end optimization of the unified forward-inverse model at the underlying algorithmic level.

4 Experiments and discussion

4.1 MR damper testing

To obtain test data for model training and validation, a performance testing platform for the MR damper was established, as shown in Fig. 6. This platform mainly has three modules: mechanical excitation, current drive, and data acquisition. For mechanical excitation, a PWD—10 electric servo shock absorber testing machine (Shandong Bangce Testing Machine Co., Ltd.) was employed to apply precise periodic displacement perturbations to the damper. It also records the transient damping force, displacement, and velocity signals synchronously. The tested object is a CSMTM0023 MR damper used in Cadillac vehicles (Zhejiang Keyi International Intelligent Suspension Technology Co., Ltd.). The current drive module utilized an NGP800 high-precision programmable DC power supply (Rohde & Schwarz GmbH). All electromechanical signals were processed via host computer software to real-time closed-loop control and high-frequency synchronous storage.

To capture the nonlinear fluid dynamic characteristics of the MR damper, basic tests used standard sinusoidal displacement excitations. The excitation amplitudes A were

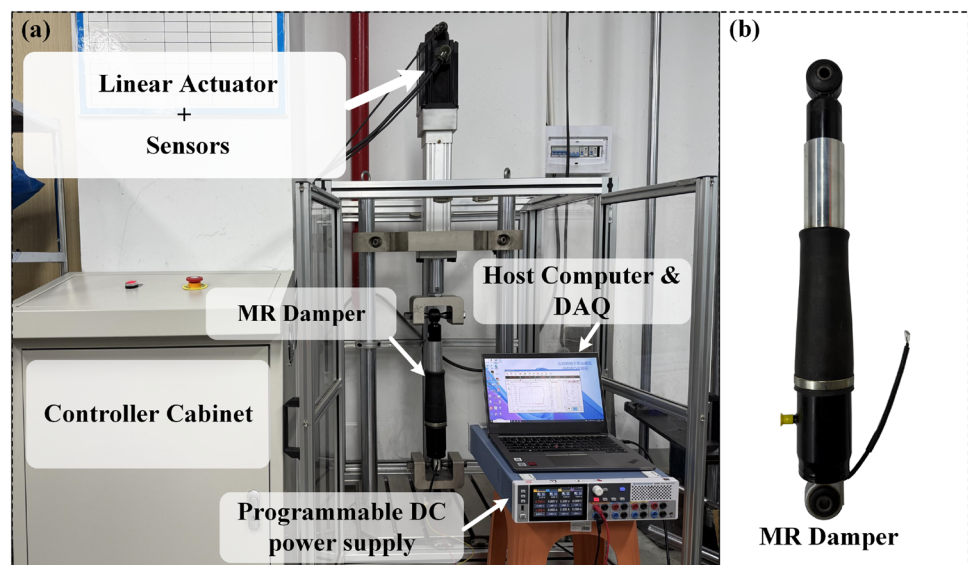
set to {5, 10, 15, 20} mm, and the frequencies f were set to {1.0, 1.5, 2.0} Hz. For electromagnetic excitation, the current I was configured at five baseline levels from 0.0 A to 2.0 A, with an increment of 0.5 A. More importantly, to comprehensively assess the zero-shot generalization ability of the PDIA under unknown conditions, an additional independent test set was collected. This set included non-standard currents and amplitudes and was solely used for subsequent generalization tests. Finally, the collected data were normalized and segmented via a sliding window, and then repackaged into datasets with a sequence length of $L = 64$.

4.2 Forward prediction and inverse deduction accuracy

The established dataset was utilized to train and test the proposed PDIA. The model construction, training, and testing were implemented using the PyTorch framework in Python 3.11. The hardware setup includes an Intel i7-12,700 CPU and an RTX 5060 GPU. Additionally, the training hyperparameters were configured as follows: batch size: 128, epochs: 300, learning rate: 0.0005, Lambda fwd: 1.0, Lambda inv: 20.0, and Epsilon: 0.0001.

To validate the comprehensive performance of the proposed PDIA in both forward prediction and inverse deduction tasks, an end-to-end accuracy evaluation was conducted on a complete test set encompassing diverse operating conditions. Initially, 500 initial sample points from the test set were selected to plot the macroscopic forward and inverse prediction curves of the PDIA, as shown in Fig. 7. The forward damping force prediction is shown in Fig. 7a, demonstrating that even under complex test conditions with continuous high-frequency

Fig. 6 MR damper testing experiment. **a** The overall testing system. **b** The MR damper under test



fluctuations, the predicted damping force closely aligns with the true data. No obvious amplitude attenuation or phase lag is observed throughout the process. This confirms the accurate fitting capability of the forward model for damping force signals. The critical inverse current deduction performance is illustrated in Fig. 7b. For the highly fluctuating target current, the inverse model achieves precise current tracking by leveraging the inherent advantages of the underlying integrated invertible structure. The predicted current exhibits no overshoot or divergence at abrupt transient edges, thereby providing strong evidence for the model's inference stability in time-domain sequences.

To further evaluate the performance of the PDIA on the entire dataset, the global accuracy and reliability of the model are assessed from a statistical perspective. Figure 8 shows the kernel density estimation (KDE) regression scatters and the absolute error spatial distributions for both forward force prediction and inverse current deduction across the whole dataset. As shown in Fig. 8a and b, the forward and inverse prediction scatters (blue high-density areas) are densely contracted and strictly aligned with the ideal prediction diagonal line $y=x$. The high-density

regions are distributed entirely along the diagonal without any obvious deviation, divergence, or local clustering. This essentially validates that the PDIA can avoid local generalization collapse. In traditional neural networks, such collapse is often induced by uneven data distribution or insufficient condition coverage when handling strongly nonlinear problems. In contrast, the PDIA can achieve unbiased and accurate predictions across all operating conditions and full scales. Meanwhile, the absolute error distributions in Fig. 8c and d further quantify the model's global stability. They reveal that up to 95% of the prediction errors for test samples are tightly bound within very low limits. Specifically, the 95% error bound for the forward damping force prediction is only 0.031 kN, and that for the inverse current deduction is merely 0.027 A.

Following the macroscopic and statistical analyses, an in-depth study was conducted using independent test results from four representative combinations of amplitude, frequency, and current excitations. As shown in the matrix plots of Fig. 9, the rows sequentially present the full-gradient test results from large-stroke/high-current to small-stroke/low-current conditions. Based on the $F-x$ and $F-v$

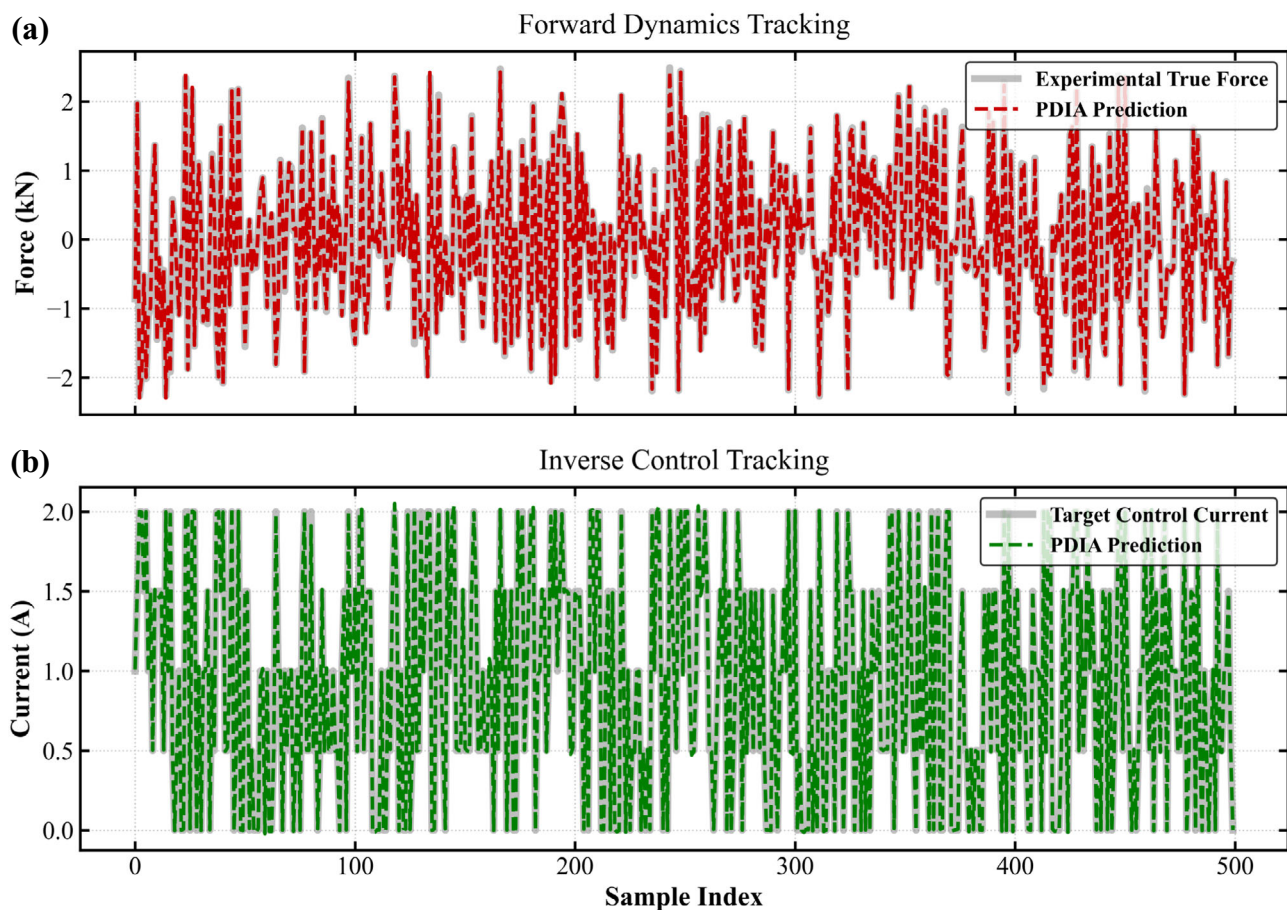


Fig. 7 Prediction performance of the PDIA on the test set. **a** Forward damping force prediction. **b** Inverse current deduction

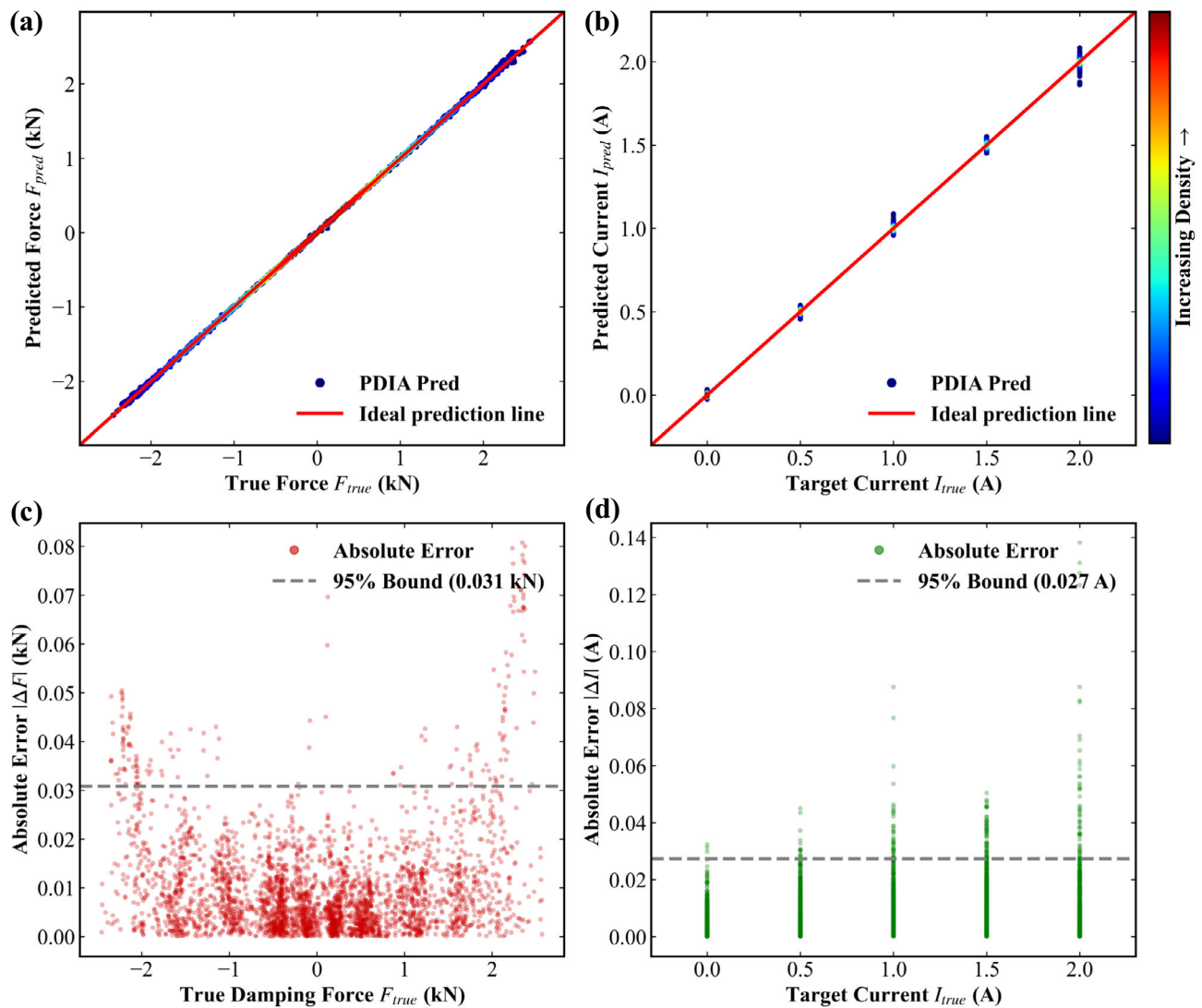


Fig. 8 Global regression analysis and absolute error distributions of the PDIA on the complete test set. **a** Forward force regression and **b** inverse current regression (scatters colored by KDE). **c** Spatial

distribution of absolute force error versus true damping force. **d** Spatial distribution of absolute current error versus target current

curves in the left and middle columns, strong nonlinear effects occur. These include the fluid inertia of the MR fluid and gas chamber compression. They become prominent as the piston's excitation velocity and displacement vary. This leads to significant distortions and asymmetric evolutions in the hysteresis trajectories. Under such extreme nonlinear conditions, the prediction curves of the PDIA exhibit no rigid deviation or phase lag, demonstrating excellent predictive capabilities. Across all operating conditions, the predicted trajectories perfectly fit all asymmetric distorted edges, with the forward prediction's root mean square error (RMSE) strictly kept within 0.014 kN. This result intuitively confirms the precise modeling capability of the PDIA for the strong nonlinear hysteretic characteristics of MR dampers.

As illustrated in the right column of Fig. 9, the inverse current deduction performance of the PDIA is further validated. As clearly illustrated by the nested secondary Y-axes and gray shaded areas, the PDIA consistently limits the peak absolute error of inverse current deduction to approximately 0.05 A, with the RMSE remaining below 0.016 A across all conditions. This high accuracy is attributed to its underlying integrated invertible structure and robust division operator. This result fully proves that the proposed model balances both exceptional accuracy and stability in the inverse current deduction task.

To further validate the generalization capability of the PDIA framework, a dedicated test set was constructed by concatenating multiple previously unseen, non-standard operating conditions in the time domain. The evaluation

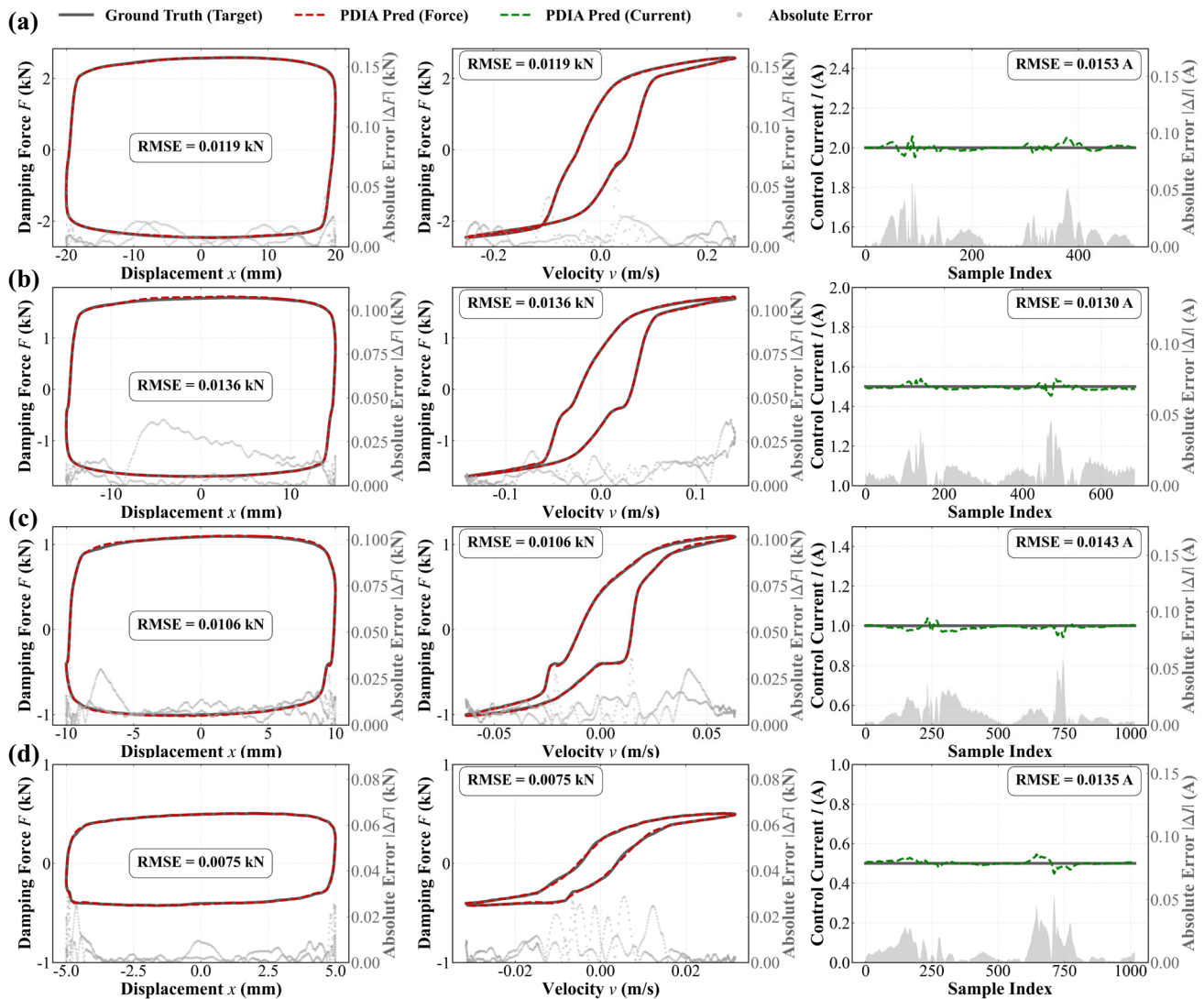


Fig. 9 Integrated forward-inverse dynamic tracking performance of the PDIA framework under multiple operating conditions. **a–d** sequentially display the independent test results under four combinations of amplitude, frequency, and target current excitations (20 mm-2 Hz-2A, 15 mm-1.5 Hz-1.5A, 10 mm-1 Hz-1A, and 5 mm-1 Hz-0.5A, respectively). The left and middle columns characterize

the forward dynamic prediction performance through F - x and F - v curves, respectively, while the right column illustrates the inverse control current deduction results. The nested Y-axes (light gray scatters and shaded areas) intuitively reflect the spatial distribution of the absolute prediction errors

results on this dataset are presented in Fig. 10. As shown in Fig. 10a, the predicted damping force of the PDIA achieves a full-time-domain match with the actual damping force on the generalization dataset. Even under sudden current step changes, the model accurately captures the microscopic evolution trajectories of the damping force. The locally magnified inset further highlights the fitting capability of the PDIA in nonlinear local distortion regions. Quantitative results indicate that the forward prediction $RMSE$ and MAE under this scenario are merely 0.019 kN and 0.015 kN, respectively. This fully verifies the strong generalization capability of the model toward unknown and variable operating conditions. Figure 10b demonstrates the

inverse current deduction performance of the PDIA in response to step currents under unseen conditions. The deduced current highly coincides with the ground-truth target trajectory. As indicated by the absolute error shaded area at the bottom, apart from extremely minor high-frequency fluctuations, the inverse deduction error across all conditions is strictly suppressed below 0.05 A. The quantitative metrics ($RMSE$ of 0.022 A and MAE of 0.018 A) firmly prove the current deduction stability and engineering reliability of the PDIA in unseen scenarios.

To quantitatively evaluate the individual contributions and relative importance of the core PDIA components, comparative ablation studies are conducted across four

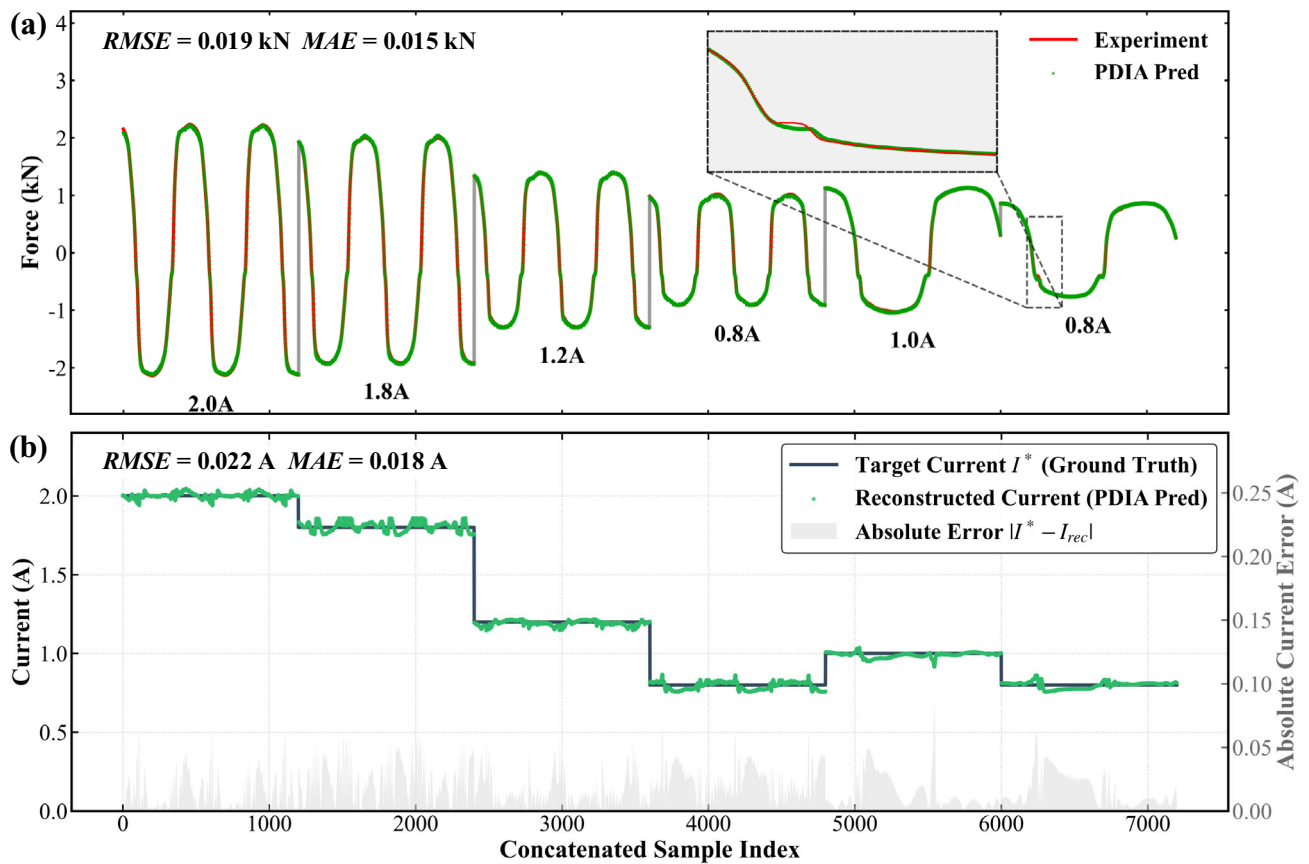


Fig. 10 Generalization performance of the PDIA framework under unseen operating conditions. **a** Time-domain tracking of the forward damping force under step-changing current excitations. **b** Raw inverse

current deduction tracking corresponding to the step target currents. The secondary right Y-axis presents the absolute current error of the inverse deduction

distinct branch configurations on the test dataset, with the results summarized in Table 1.

As shown in Table 1, the proposed full PDIA achieves the highest tracking accuracy across all forward and inverse metrics. Disabling the monotonicity constraint (Branch a) causes the inverse current RMSE to increase significantly to 1.6551A, demonstrating that the absence of strict monotonicity introduces non-physical multi-valued mappings that disrupt bisection search convergence. Deactivating the robust division operator (Branch b) spikes the inverse current RMSE to 3.0148 A and the zero-crossing error to 2.0798. This indicates that near the velocity zero-crossing points (where the motion state modulation factor approaches zero), standard direct division triggers division-by-zero singularities that severely amplify high-frequency measurement and prediction noises. Furthermore, removing the bidirectional latent-space alignment (Branch c) elevates the inverse current RMSE to 2.1682A, confirming that joint bidirectional optimization is indispensable to calibrate the parameters of the inverse state observer. Based on the magnitude of error increases under ablated states, the relative importance of

the components to the overall inverse performance is ranked as: the robust division operator, followed by the latent-space alignment strategy and the monotonicity constraint. These findings verify that all proposed components are essential and work synergistically to ensure numerical stability and tracking accuracy.

4.3 Architecture analysis and closed-loop consistency verification

This section evaluates the physical authenticity of the internal features within the PDIA and verifies the closed-loop consistency of the unified forward-inverse modeling. In-depth validations are conducted across three dimensions: internal latent variable analysis, bi-directional closed-loop verification, and long-term autoregressive simulation.

To intuitively verify the alignment between the actual network performance and the design objectives, a visual decoupling analysis is performed on the three core latent variables: the base dynamic bias B , the state modulation factor S , and the electromagnetic core G . This analysis was

Table 1 Quantitative metrics of the ablation studies

Ablation branch configuration	Forward		Inverse		Zero-crossing error ($ v_i < 0.005$)
	<i>RMSE</i>	<i>MAE</i>	<i>RMSE</i>	<i>MAE</i>	
Full PDIA (Proposed)	0.0174	0.0123	0.0248	0.0188	0.0202
(a) No monotonicity	0.0206	0.0156	1.6551	1.5220	1.2441
(b) No robust division operator	0.0185	0.0131	3.0148	2.8837	2.0798
(c) No latent alignment	0.0172	0.0126	2.1682	1.8363	1.9813

conducted under multiple operating conditions with sequentially decreasing currents and velocities, with the results illustrated in Fig. 11.

The network then autonomously reconstructs the base dynamic bias B (the second column in Fig. 11). This variable accurately restores the passive mechanical dissipation hysteresis loop. This loop successfully couples structural friction, fluid viscosity, and gas chamber stiffness, with its amplitude decaying synchronously as the piston velocity decreases. The resulting smooth, distortion-free trajectory confirms the model's precise capability to extract underlying physical nonlinearities.

The state modulation factor S (the third column of Fig. 11) is strictly constrained by the energy dissipation criterion ($S \cdot v \geq 0$) and the symmetric tanh activation. Consequently, it autonomously manifests as a smooth, hysteretic sign curve in the latent space that traverses the positive and negative boundaries. This curve not only captures the nonlinear microscopic evolution of the MR fluid but also compensates for intricate dynamic hysteresis details, thereby significantly improving the overall fitting accuracy.

In the electromagnetic response dimension (the fourth column of Fig. 11), the extracted core mapping $G(I)$ strictly satisfies the physical constraints. By precisely passing through the physical origin ($G(0)=0$) and maintaining a strict monotonic increase across the entire domain ($dG/dI > 0$), it establishes a monotonic bijective mapping between the control current and the damping force. This structural design completely eradicates the multi-valued mapping singularity, ensuring a unique solution for the inverse current deduction.

In summary, the three core latent variables extracted by the PDIA align perfectly with the predefined design objectives and successfully satisfy all imposed physical constraints. Their evolutionary laws are highly self-consistent with the underlying physical mechanisms, which fully validates the physical interpretability and intrinsic consistency of the proposed unified forward-inverse modeling framework.

To further evaluate the forward-inverse consistency of the PDIA framework, two bi-directional closed-loop tests were implemented, tracking both the current-to-force and force-to-current reconstruction pathways. By calculating

the *MAE* and *RMSE*, the bi-directional mapping consistency of the model was quantitatively evaluated from two dimensions: electromagnetic excitation intensity and kinematic amplitude. The results are presented in Fig. 12.

As shown in Fig. 12a and b, under current excitations from 0 A to 2.0 A, the bi-directional closed-loop reconstruction errors of the PDIA architecture are strictly constrained at extremely low levels. The *MAE* of the current closed-loop reconstruction remains below 0.018 A, and the *RMSE* is stably controlled within 0.023 A. Similarly, the *MAE* of the force closed-loop reconstruction is consistently below 0.009 kN, with the *RMSE* under 0.0125 kN. Furthermore, the errors exhibit only a minor and bounded linear growth as the current increases, showing no signs of divergence.

As illustrated in Fig. 12c and d, the bi-directional closed-loop errors also maintain excellent stability under kinematic amplitudes ranging from 5 to 20 mm. The maximum error for the closed-loop current reconstruction occurs under the 10 mm condition, with an *MAE* of 0.034 A and an *RMSE* of 0.041 A. Correspondingly, the maximum error for the closed-loop force reconstruction also appears under this condition, yielding an *MAE* of 0.0072 kN and an *RMSE* of 0.0088 kN. These results conclusively demonstrate that the PDIA architecture successfully fulfills its closed-loop design objectives. The bi-directional closed-loop errors are tightly confined within minimal bounds, verifying the high consistency of the forward-inverse mapping.

To validate the temporal dynamic stability of the PDIA architecture, a long-term hybrid autoregressive simulation without external data correction was designed. In this experiment, the model operates completely independent of external real current data, relying instead on self-regression. Specifically, the inverse model generates command currents based on the desired force, which then serve as inputs for the forward dynamic model to drive the system evolution in the subsequent time step. The results over 1500 time steps are presented in Fig. 13.

As shown in Fig. 13a, the inverse controller exhibits exceptional numerical stability when responding to the target force trajectory. The inferred command current highly matches the target current, yielding a full-process *RMSE* of only 0.0141 A. Moreover, the generated current

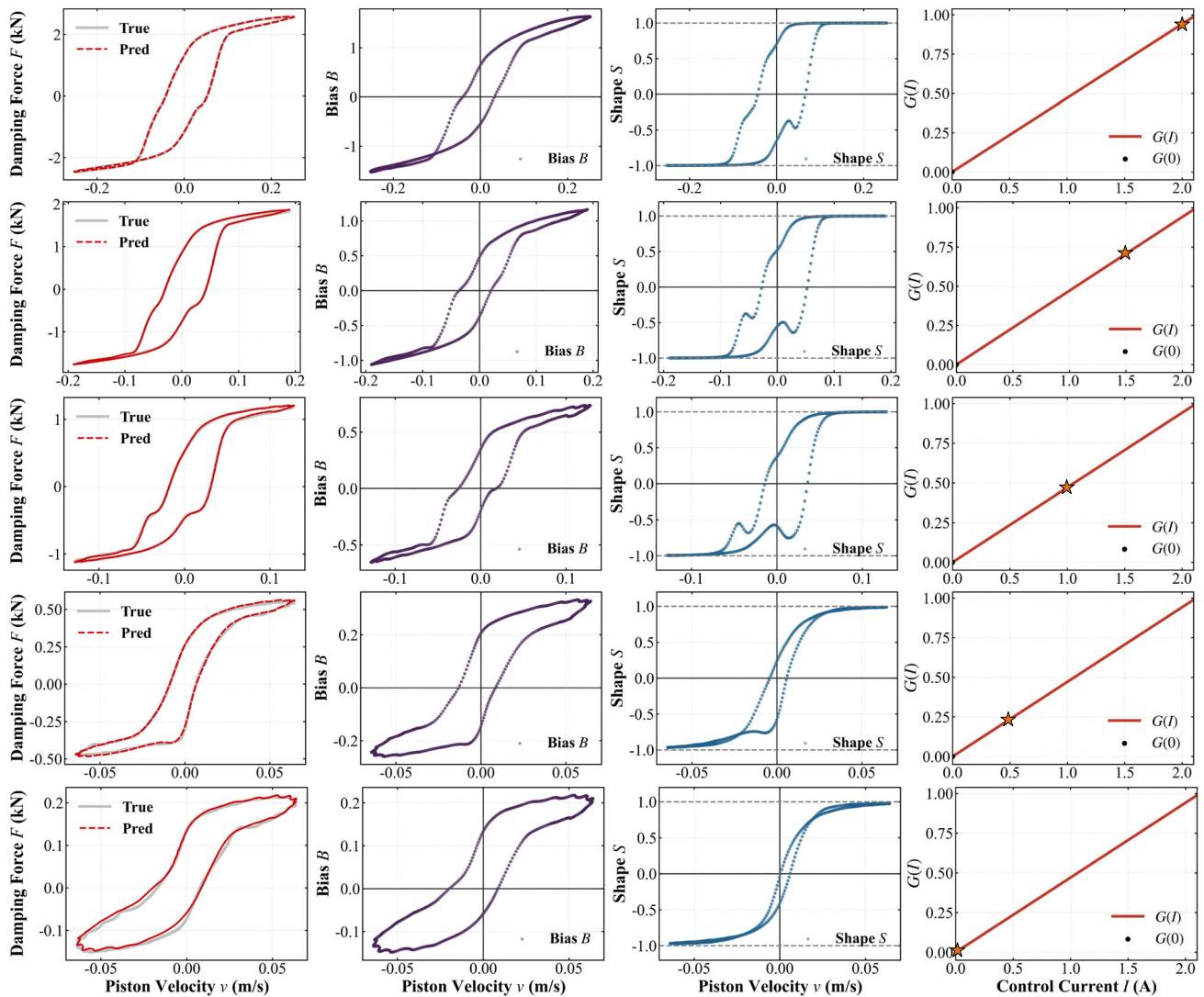


Fig. 11 Visual perspective of internal variables in the PDIA architecture under multiple operating conditions. From top to bottom, the rows represent test conditions with sequentially decreasing current excitations and kinematic velocities. The columns (from left to right) display: the macroscopic F - v tracking loops, the micro-decoupled

base dynamics B , the motion state modulation S , and the strictly monotonic electromagnetic core $G(I)$. The star markers (★) precisely locate the static equivalent magnetization operating points corresponding to each current condition

command curve is remarkably smooth, without obvious numerical jumps or high-frequency oscillations. As shown in Fig. 13b, in autoregressive simulations, any minor inversion error theoretically accumulates over time due to recursion. However, the reconstructed force generated by the PDIA exhibits no obvious phase shift or amplitude divergence over three complete cycles, with the $RMSE$ stabilizing at 0.0145 kN. In summary, during continuous closed-loop recursive deduction, the model consistently maintains excellent performance characterized by low noise. The errors do not accumulate or amplify over time,

fully confirming the strong stability and robustness of the PDIA.

4.4 Comparative study

To comprehensively evaluate the performance of the proposed PDIA, parametric models (Hyperbolic Tangent, RC Operator, and Modified Bouc-Wen) alongside representative black-box and physics-informed neural networks (FNN, RNN, NLSTM, and PINN) were selected as baselines for the multi-dimensional comparative assessment.

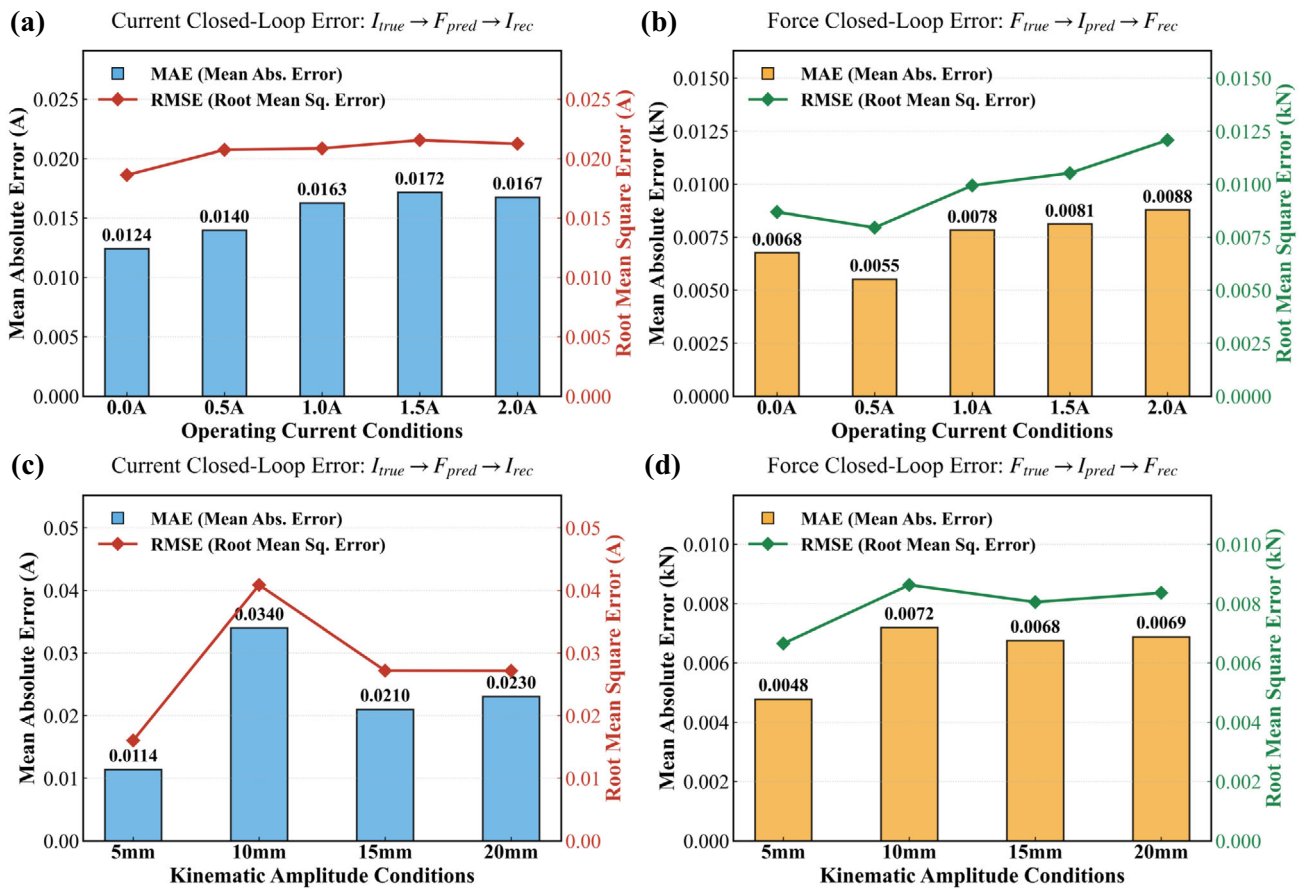


Fig. 12 Bi-directional closed-loop consistency verification of forward force prediction and inverse current deduction. The bar charts (left Y-axis) and line plots (right Y-axis) represent the MAE and RMSE, respectively. **a** and **b** show the closed-loop current and damping force

reconstruction errors under different electromagnetic excitations (0 A to 2.0 A). **c** and **d** display the closed-loop errors under varying kinematic amplitudes (5 mm to 20 mm)

First, a performance comparison between the PDIA and parametric models was conducted on the same 2.0 A operating condition dataset. The damping force prediction results for each model are shown in Fig. 14. Based on the macroscopic trajectories, the identified parametric models can adequately fit the basic hysteresis contour of the damper. However, restricted by their rigid parameterized structures, these models exhibit pronounced errors in the peak regions when attempting to capture the asymmetric features of the real damper. In contrast, the proposed PDIA architecture seamlessly integrates the underlying physical skeleton with the universal approximation capability of deep learning. Consequently, the network precisely reproduces the authentic local micro-distortions without any apparent deviation.

To provide a more rigorous comparison, the kernel density estimation (KDE) regression of the forward predicted force for each model was computed, as detailed in Fig. 15. Under an ideal mapping, the scattered predictions should strictly converge along the red diagonal line. As shown in Fig. 15a-c), due to the fitting errors of the

parametric models at the time-domain peaks, the baseline models exhibit varying degrees of trajectory twisting and deviation in the strongly nonlinear hysteresis regions, causing the predicted forces to deviate from the ideal diagonal. Quantitative results indicate that the forward prediction RMSE of the Hyperbolic Tangent, RC Operator, and Bouc-Wen models are 0.1472 kN, 0.1616 kN, and 0.1751 kN, respectively. In sharp contrast, as shown in Fig. 15d, the damping force predictions generated by the PDIA are uniformly and tightly distributed along the diagonal, lacking any distortion or systematic offset. The forward prediction RMSE of the PDIA was reduced to merely 0.0140 kN.

Figure 16 summarizes the final comparative results via a performance radar chart. In the radar chart, the polygon's coverage area directly maps the degree of uncertainty divergence of the model in global space. Because parametric models fail to capture complex distortions, their error polygons expand outward. Conversely, the error polygon for the PDIA remains tightly clustered near the origin, indicating its minimal prediction error. These

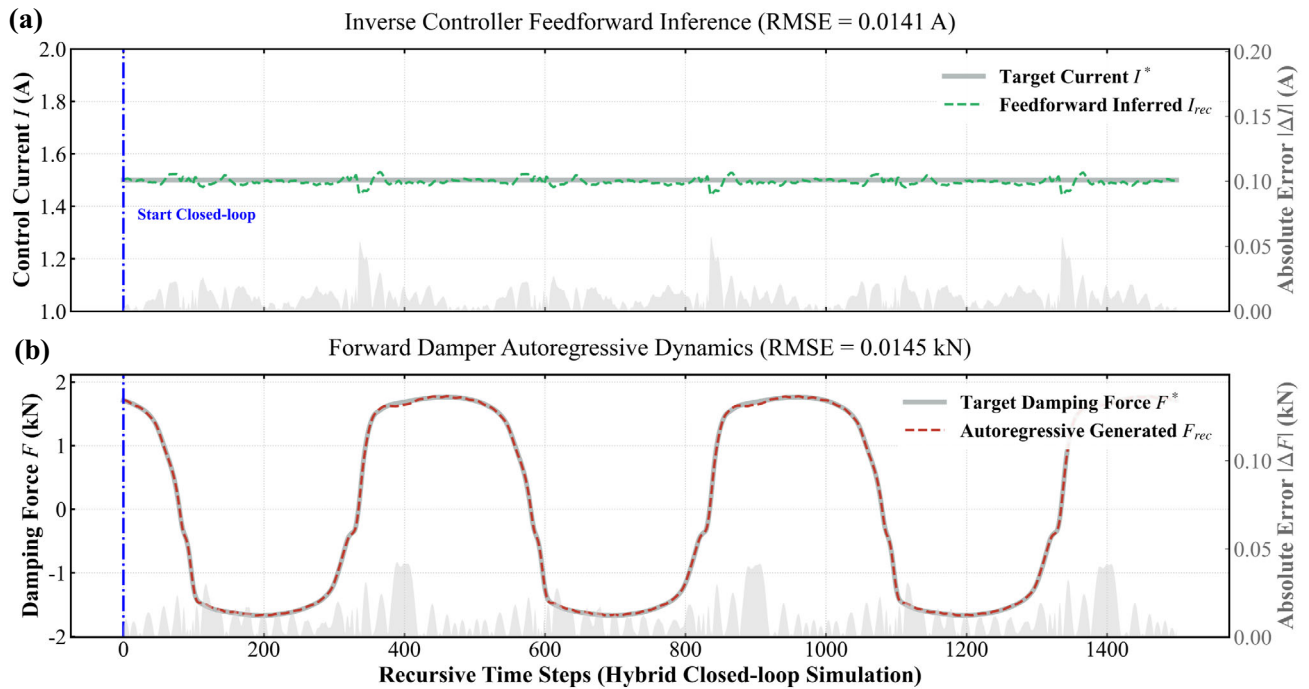


Fig. 13 Long-term hybrid autoregressive closed-loop simulation of the PDIA framework. **a** Current response curve. **b** Damping force response curve

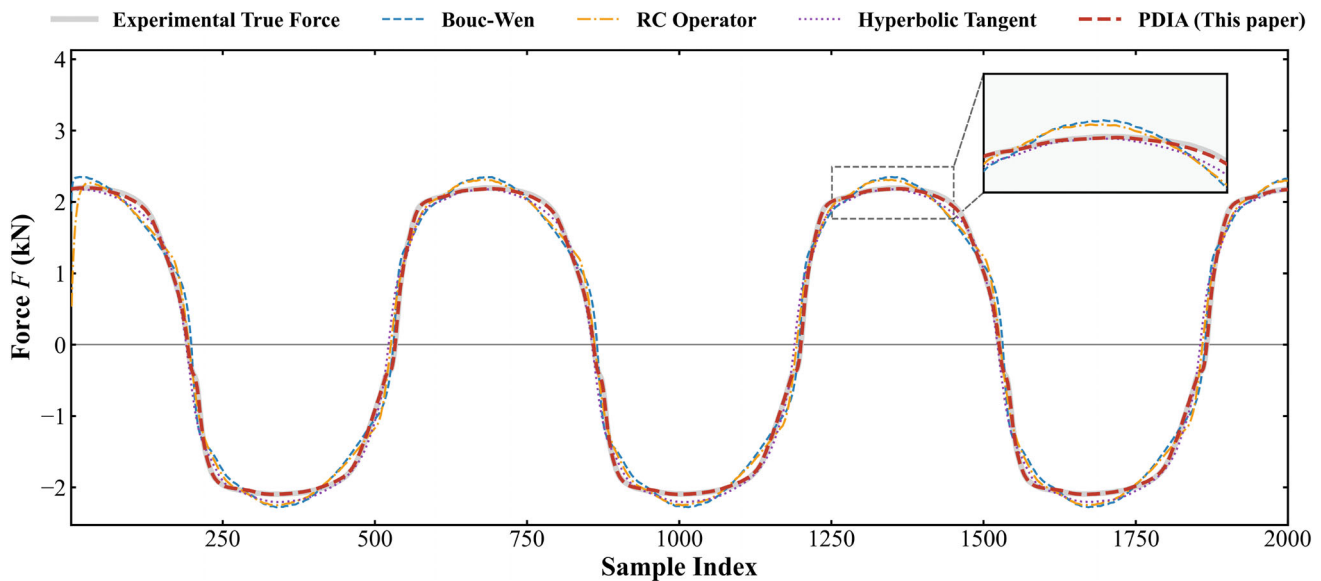


Fig. 14 Time-domain comparison of traditional models

results demonstrate that the PDIA overcomes the inherent limitations of traditional hysteresis models under strong nonlinearity. It not only possesses high physical interpretability but also fundamentally ensures high-fidelity forward force prediction.

Following the evaluation of the parametric models, a memoryless feedforward neural network (FNN), a physics-informed neural network (PINN) incorporating soft

physical residual constraints, a recurrent neural network (RNN), and a nested long short-term memory network (NLSTM) were selected for further comparative experiments. The modeling capabilities of these models were evaluated on the same operating condition dataset, with the results presented in Fig. 17.

Figure 17a and b display the F - x and F - v curves of the forward damping force prediction for each model.

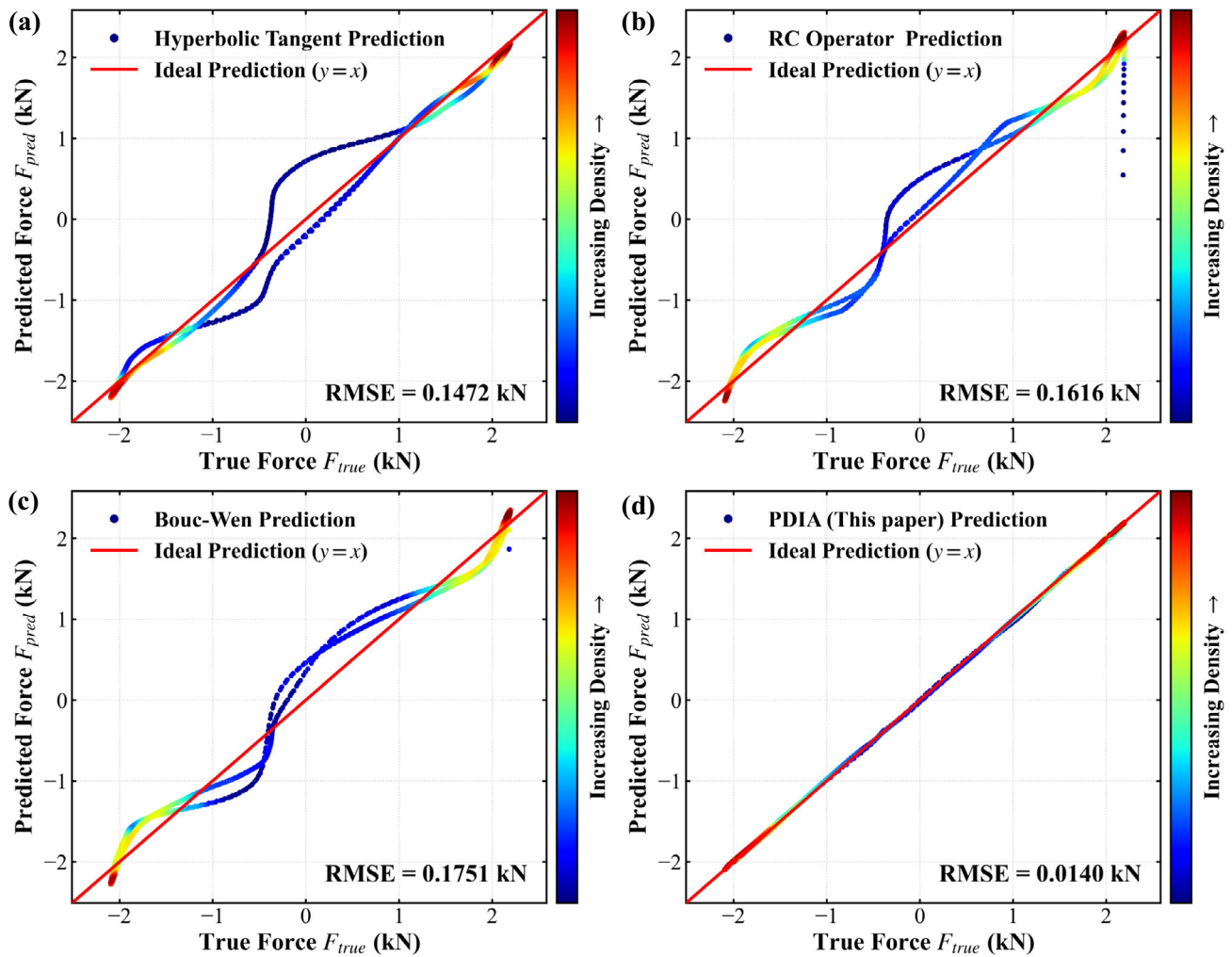


Fig. 15 Forward force regression plots of the comparative models. **a** Hyperbolic Tangent model. **b** RC Operator model. **c** Bouc-Wen model. **d** PDIA model

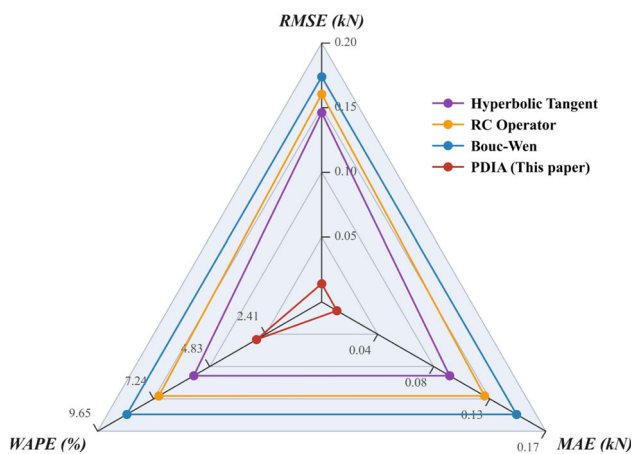


Fig. 16 Performance radar chart of the parametric models

Macroscopically, all models perform reasonably well, with only the FNN and PINN exhibiting trajectory deviations in the hysteresis regions. Figure 17c and d provide locally magnified views of the peak areas, revealing that minor deviations from the ground truth remain for the baseline models. Overall, the fitting performance ranks from lowest to highest as: FNN < PINN < RNN < NLSTM < PDIA. Analysis indicates that the FNN and PINN, lacking temporal memory mechanisms, show pronounced fitting deviations in the yield transition zones. The RNN and NLSTM, equipped with hidden state memory, perform better; however, due to the lack of explicit physical structural decoupling, they still exhibit noticeable deviations from the true trajectories at local extrema. Compared to other models, the proposed PDIA demonstrates highly

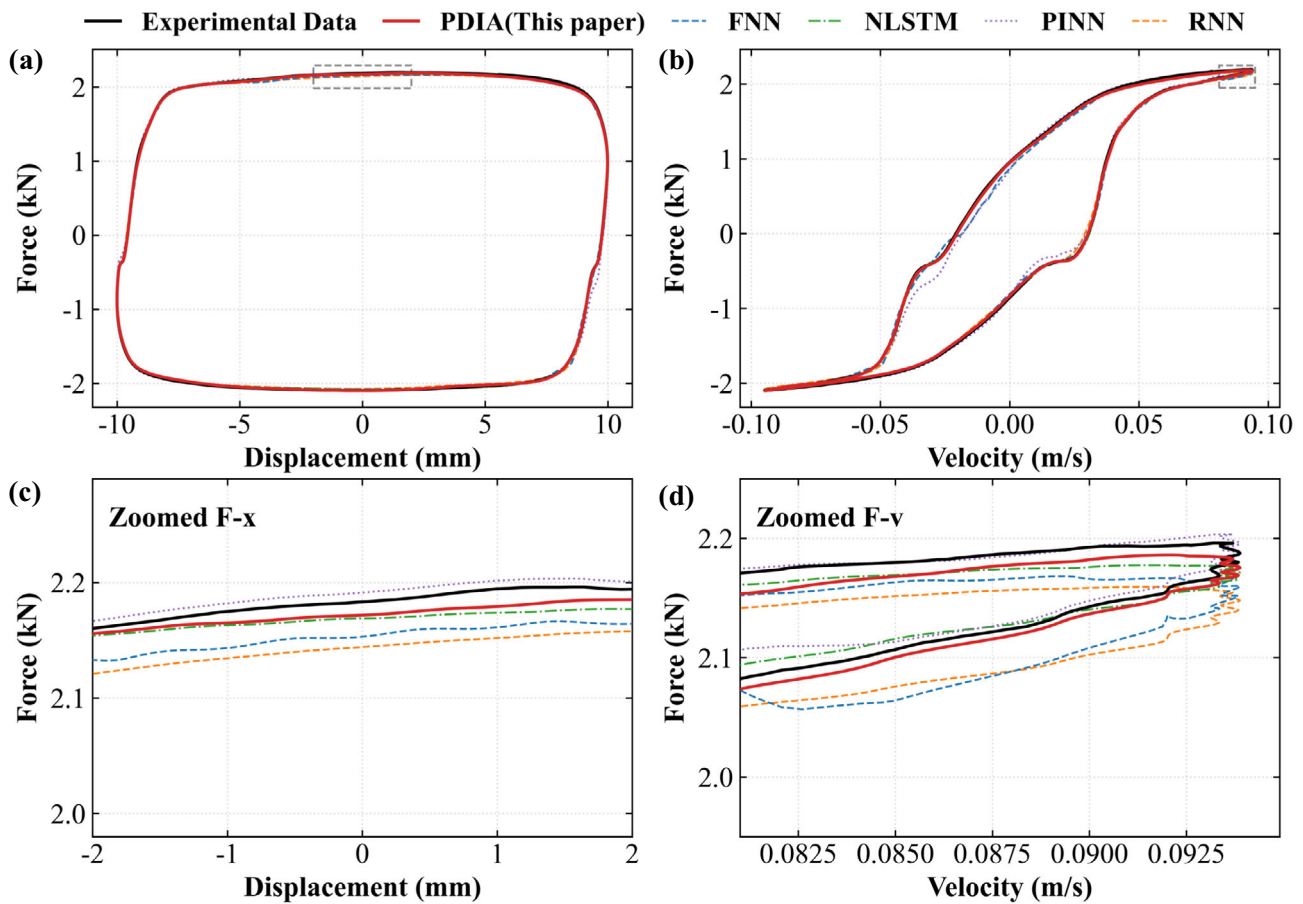


Fig. 17 Forward damping force prediction results on the same operating condition dataset. **a** F - x comparison. **b** F - v comparison. **c** Locally magnified F - x plot. **d** Locally magnified F - v plot

balanced performance across all regions, highlighting its outstanding fitting capability in the forward prediction task.

Figure 18 further compares the performance of each model in the inverse current deduction task. As shown in Fig. 18a, under stable target current commands, the memoryless FNN and PINN exhibit distinct periodic downward pulse errors. As shown in Fig. 18b, although the RNN and NLSTM avoid extreme divergence, they still suffer from continuous minor fluctuations, and their overall prediction baselines exhibit a noticeable steady-state offset. The proposed PDIA also shows slight fluctuations, but their amplitudes are significantly smaller than those of the other models, and its baseline closely tracks the target current. For a more in-depth analysis, error scatter distributions and low-velocity violin plots were generated, as depicted in Fig. 19.

As shown in Fig. 19a, the memoryless FNN and the soft-constrained PINN produce severely divergent error scatters in the low-velocity regions on both sides. Although the RNN and NLSTM (with temporal memory) suppress this divergence to some extent, they still exhibit error

bulging and deviation in the low-velocity zone due to the lack of rigid underlying mathematical constraints. In contrast, the absolute error of the PDIA is consistently suppressed within an extremely low interval across the entire velocity spectrum. This result clearly highlights the inherent limitations of traditional neural networks in multi-valued mapping areas. The low-velocity violin plot in Fig. 19b further displays the statistical error distribution of each model. The absolute errors of the FNN and PINN exhibit a long-tailed distribution. The RNN and NLSTM perform relatively stably in the low-velocity zone, but a distinct offset exists between their error distributions and the zero baseline. Conversely, the error probability density of the PDIA is highly concentrated near the zero baseline, indicating superior inverse deduction accuracy compared to other models.

To examine the model's robustness outside the training distribution, evaluations were conducted on a generalization dataset comprising unseen operating conditions. The comparison of inverse current deduction results is shown in Fig. 20, and the bi-directional evaluation metrics are

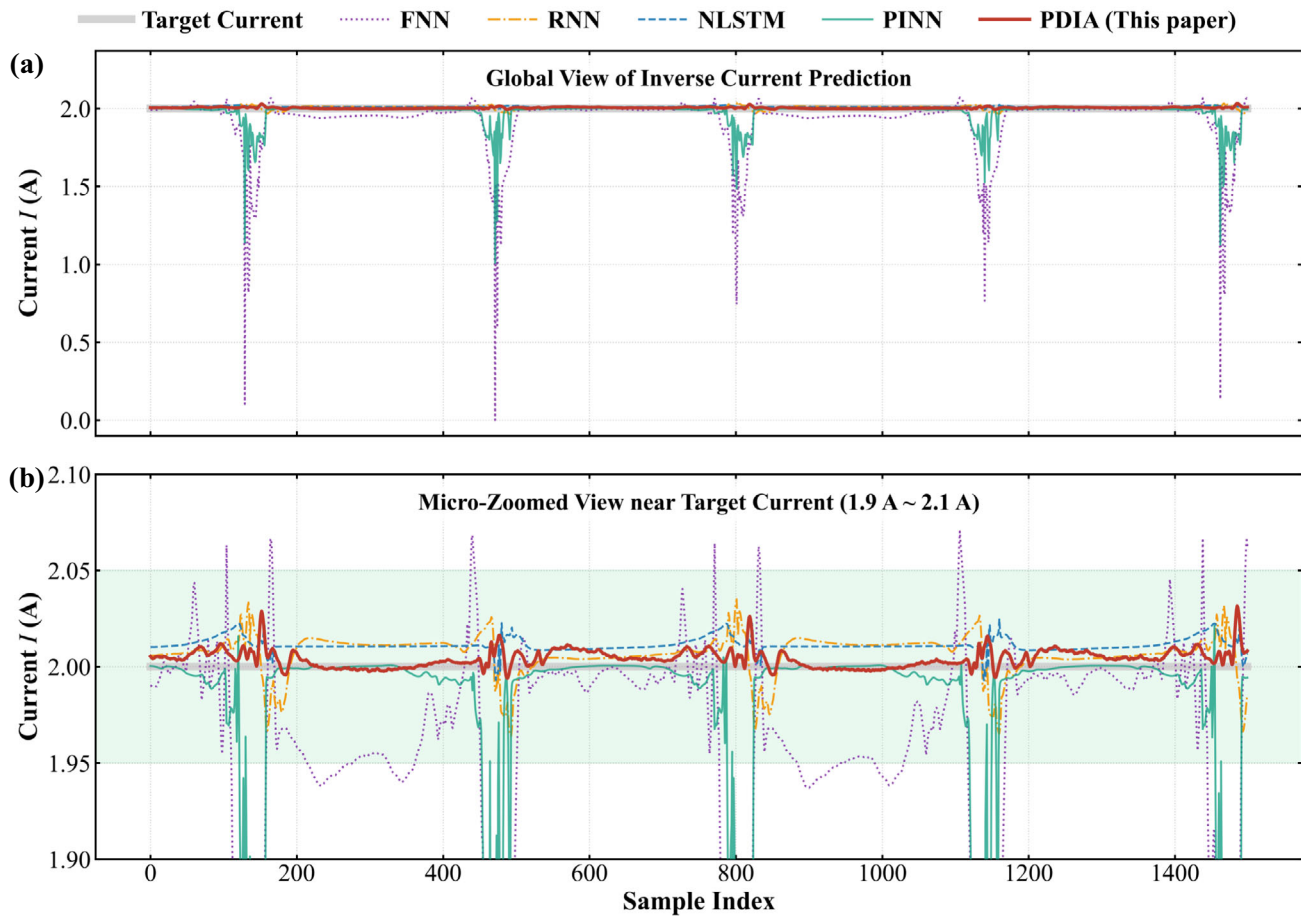


Fig. 18 Inverse current deduction results on the same operating condition dataset. **a** Overall comparison. **b** Locally magnified comparison

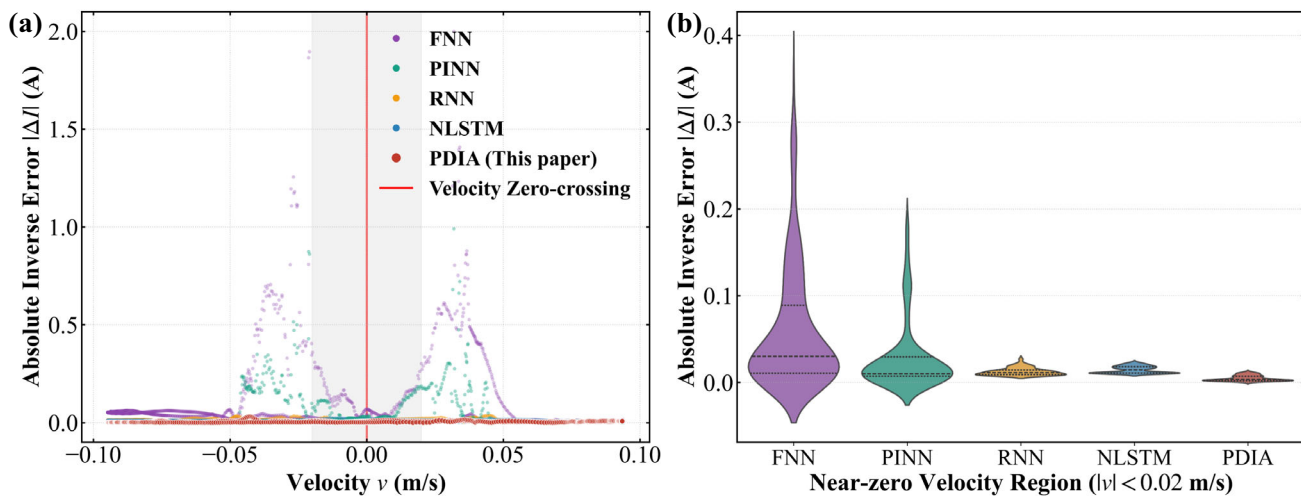


Fig. 19 Inverse current deduction errors. **a** Error scatter distribution. **b** Violin plot in the low-velocity region

summarized in Table 2. In forward dynamic generalization (the left side of Table 2), the accuracy ranking of the models remains largely consistent with the in-distribution scenario. However, in the inverse current deduction

generalization (the right side of Table 2), the models exhibit different error distribution characteristics.

The PDIA achieves the optimal generalization performance, with its inverse $RMSE$ and other metrics being the

lowest in the group. Although the current deduction curve shown in Fig. 20 has slight fluctuations, it closely tracks the target current across the entire time history, proving the reliable generalization of the hard-constrained topology in unknown physical domains. The PINN ranks second; although Fig. 20a shows that the PINN still has severe downward spikes, its overall *RMSE* is second only to the PDIA. This indicates that the soft penalty term in the loss function helps suppress the macroscopic baseline drift during inference in unknown domains.

Interestingly, the FNN yields the highest *RMSE* in the group, yet its *MAE* and *WAPE* are unexpectedly lower than those of the RNN and NLSTM. This occurs because the *RMSE* penalizes local extreme outliers more heavily, whereas the *MAE* and *WAPE* better reflect the global average deviation. The transient pulse errors of the FNN inflate its *RMSE* metric; however, since no global baseline drift occurs, its average error remains relatively controlled. Conversely, for the sequence models containing hidden states (RNN and NLSTM), the evolution of their hidden state feature vectors is highly susceptible to drift when processing unseen continuous motion sequences. Although this drift does not trigger extreme local spikes, it significantly increases the global steady-state error, ultimately causing its *MAE* and *WAPE* metrics to be inferior to those of the FNN.

Table 2 Forward and inverse prediction performance on the generalization dataset

Model	Forward performance			Inverse performance		
	<i>RMSE</i>	<i>MAE</i>	<i>WAPE</i>	<i>RMSE</i>	<i>MAE</i>	<i>WAPE</i>
FNN	0.2184	0.0924	7.5525	0.1907	0.0892	6.5776
PINN	0.1434	0.0660	5.3908	0.1158	0.0774	5.6914
NLSTM	0.0577	0.0418	3.4169	0.1476	0.1151	8.4629
RNN	0.0799	0.0513	4.1945	0.1323	0.0970	7.1353
PDIA	0.0239	0.0170	1.3927	0.0304	0.0211	1.7494

Bold values are adopted to highlight the prediction performance of our proposed PDIA model

Based on the above in-depth analysis, memoryless networks are highly susceptible to local multi-valued mapping interference, triggering severe spikes. Meanwhile, time-series networks are prone to long-term steady-state distortion caused by hidden state drift during generalization extrapolation. More fundamentally, the PINN, which attempts to append physical equation residuals solely to the loss function, is equally prone to triggering spikes. This indirect intervention mechanism based on gradient optimization lacks mandatory constraints in the underlying network; thus, physical constraints easily fail when encountering unseen data. The proposed PDIA architecture discards soft penalties. Through underlying physical hard

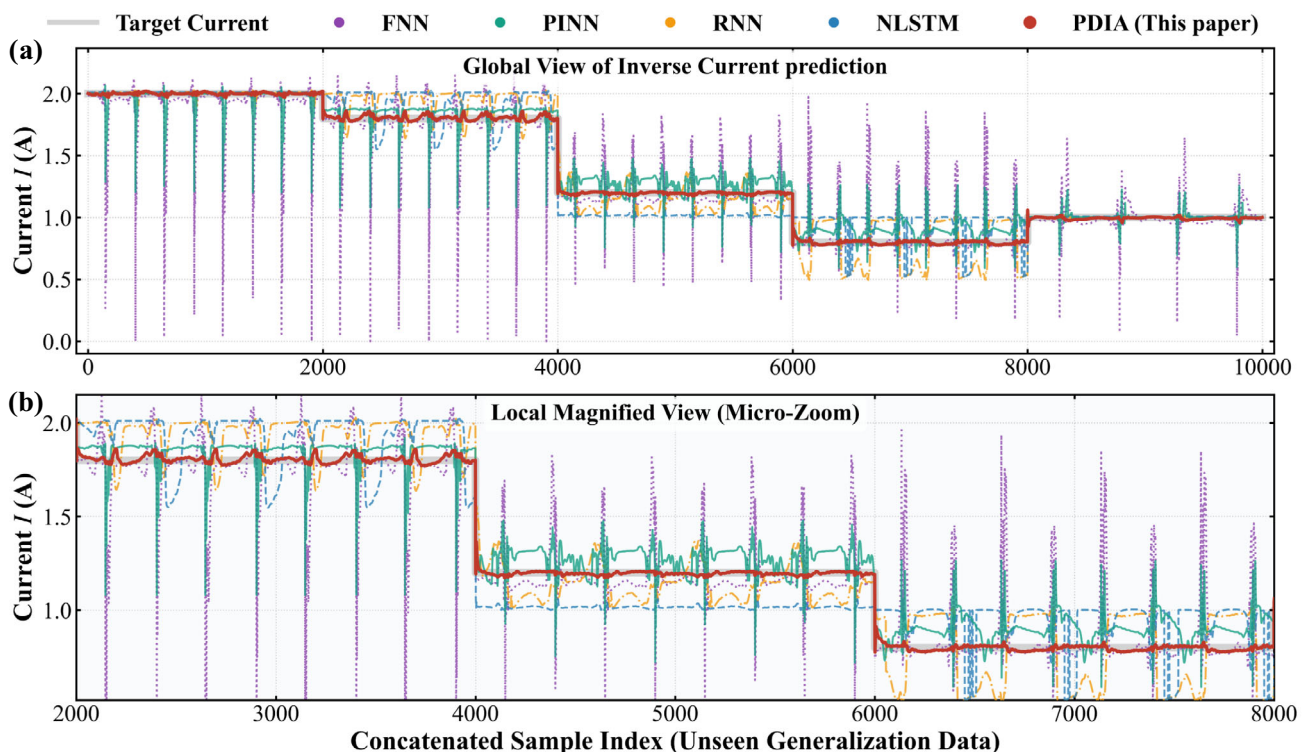


Fig. 20 Inverse current deduction results in the generalization experiment. **a** Overall comparison. **b** Locally magnified comparison

constraints and macroscopic structural decoupling, it effectively balances the continuity of temporal evolution with the certainty of inverse solving, achieving a unification of accuracy and stability under generalized operating conditions.

To further evaluate the multi-dimensional performance of the proposed model under generalization excitation conditions, three recent advanced models are also selected for comparative experiments: the Hammerstein model, the MEA-BP model, and the TF-BPNN model. The evaluation results are normalized to the range of [0.2, 1.0], and the results are shown in Fig. 21 (performance radar charts); a smaller polygon area represents lower errors and superior performance (closer to the center value of 0.2 represents the optimum).

As illustrated in Fig. 21a for the forward dynamics task, the proposed PDIA forms the tightest, innermost triangle at the 0.2 core across all three axes, establishing a narrow error envelope. Conversely, Hammerstein expands outwards because its linear dynamic block faces challenges in representing the non-linear hysteretic behavior. MEA-BP stretches furthest outwards, especially along the WAPE axis, because it operates as an unconstrained black-box, making it prone to prediction fluctuations. For the inverse task shown in Fig. 21b, the PDIA again maintains the most compact innermost triangle at the 0.2 center. In contrast, TF-BPNN exhibits an asymmetrical polygon with a massive outward spike on the RMSE axis, which visually illustrates the numerical sensitivity and current tracking fluctuations it experiences near zero-velocity regions due to the lack of division regularization. By contrast, the PDIA

incorporates a robust division operator with a stabilizing factor and topologically enforces strict monotonicity, ensuring both numerical stability near zero-crossings and superior bidirectional accuracy.

4.5 Force tracking validation

To comprehensively evaluate the PDIA, a force tracking experiment was conducted using the aforementioned test-bed. In this experiment, the host computer generated a pseudo-random desired force trajectory featuring multi-frequency superposition and time-varying amplitudes. The inverse model then deduces the target control current sequence. Subsequently, a high-precision programmable DC power supply outputs the target current to drive the real MR damper. Meanwhile, the testing machine provided kinematic excitations and collected the resulting damping force data. The resulting damping force output curves are shown in Fig. 22.

Figure 22 illustrates the dynamic force tracking performance. At the macroscopic level, the actual output force driven by the inferred current effectively tracks the random fluctuations of the target force. The overall trajectories of both forces exhibit high consistency in phase and amplitude. This effectively verifies the accuracy of the PDIA in forward-inverse physical mapping and its feasibility for application in practical semi-active control systems.

However, as observed from the absolute error curves, local error spikes are still present during the tracking process. Specifically, these transient deviations are primarily concentrated in the peak and trough regions of the damping

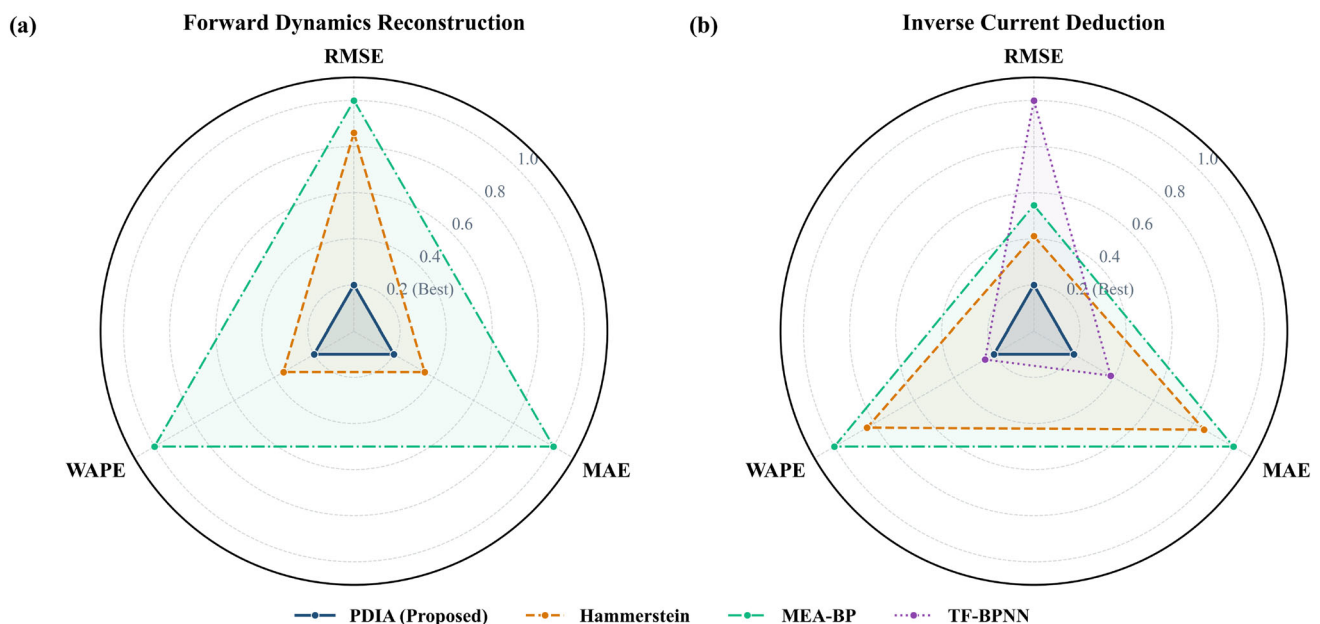


Fig. 21 Evaluation of generalization performance of different advanced models. **a** Forward force prediction. **b** Reverse current deduction

force, as well as at the instants of high-frequency sudden changes in the target commands (e.g., around $t \approx 8.5$ s in Fig. 22c). These tracking deviations mainly originate from the objective physical delay characteristics inherent in actual electromechanical coupled systems, such as the output current lag of the DC power supply. Despite the aforementioned local deviations caused by hardware physical limits, the tracking errors under the overall test conditions are constrained within a reasonable range. The experimental results further confirm that the PDIA possesses good stability and reliability in real-world deployment. They also demonstrate the significant application potential of the proposed PDIA in the semi-active control of MR dampers.

To fully validate the practical deployment capability of the PDIA inverse model, quantitative evaluations are conducted across three dimensions: parameter scale, single-sample FLOPs, and PC-side inference latency. The results show that the model contains 150,402 trainable parameters, requires a computational cost of approximately 19.448 MFLOPs (including the repeated evaluations required by the bisection procedure), and achieves an average CPU inverse inference latency of 8.164 ms. These findings indicate a compact model scale and moderate computational overhead with millisecond-level latency, demonstrating great potential for stable deployment on embedded hardware and hardware-in-the-loop (HIL) platforms through future pruning and compilation optimizations.

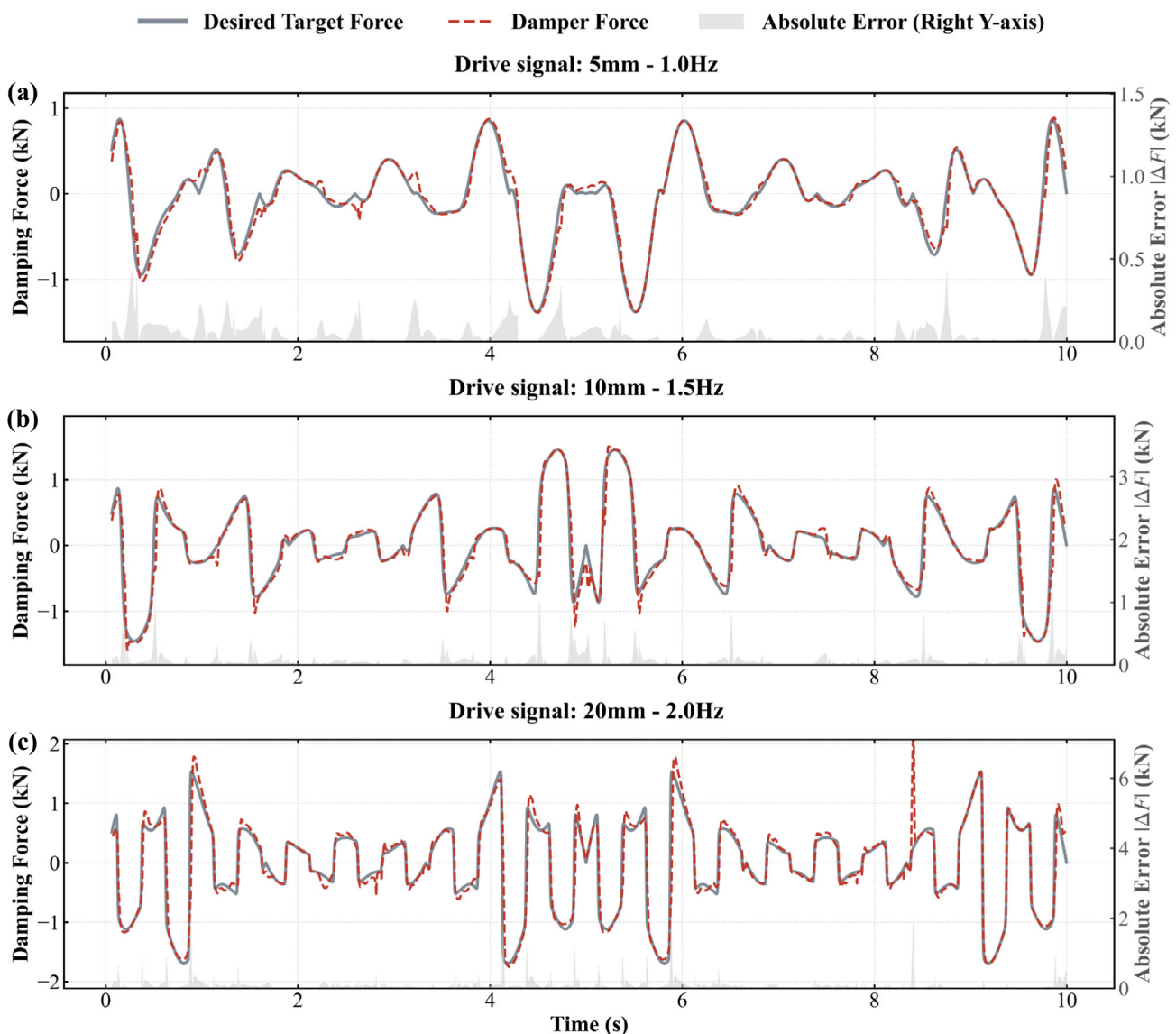


Fig. 22 Force tracking experiment results. **a** 5 mm-1.0 Hz. **b** 10 mm-1.5 Hz. **c** 10 mm-2.0 Hz

4.6 Discussion on model boundaries and limitations

As demonstrated by the preceding experimental results and analyses, the effectiveness of the proposed model has been validated. This section analyzes the operating conditions of the model and the corresponding potential application limitations. On this basis, the boundary breakdowns of the model under extreme regimes and its limitations in future practical engineering deployment are further discussed.

4.6.1 Potential limitations

Several physical assumptions are established in the PDIA model based on physical considerations. Although these assumptions are necessary conditions for achieving real-time model-invertible calculations, analyzing their applicability boundaries and corresponding potential limitations under practical operating conditions remains crucial. The detailed analysis is as follows:

- a. Health status of the MR fluid: The proposed PDIA is trained on a healthy, well-dispersed fluid. In long-term service, chemical aging, mechanical shear, and particle sedimentation inevitably reduce the fluid's effective yield stress and base viscosity. This degradation introduces a persistent bias in both force prediction and inverse current deduction. While this simplification is valid within the standard fluid service life, severe degradation necessitates periodic parameter recalibration or offline aging compensation.
- b. Thermal drift and temperature fluctuations: In practical applications, temperature sensors are omitted to minimize system complexity and hardware costs. Although the training dataset includes temperature rise processes, active thermal feedback is not coupled into the model. Under continuous high-current operations, heat buildup alters coil resistance and carrier oil viscosity. Without real-time thermal compensation, this thermal drift causes a gradual, minor offset in force tracking and current deduction, which is expected to be managed by the system's closed-loop feedback controller.
- c. Magnetic hysteresis and remanence: The damper's overall hysteresis is unified and predicted within the state observer, encompassing the magnetic hysteresis of the core, while the invertible core assumes strict current-to-force monotonicity. Despite successful hysteresis modeling, minor tracking errors or dynamic delays may occur during extremely rapid transient current changes. This stems from the discrepancy between the static-monotonic assumption and the high-frequency, path-dependent remanence of the steel core. Under normal operating frequencies, this error behaves

as bounded high-frequency noise that does not compromise closed-loop stability.

When the operating conditions of the aforementioned three aspects degrade further under extreme scenarios, they will evolve into severe structural breakdowns, specifically manifesting as total MR fluid failure (where fluid failure completely alters the baseline rheology of the fluid), extreme thermal runaway (where high temperature collapses carrier fluid viscosity and deforms the monotonic current-to-force relationship), and severe magnetic hysteresis lag. Beyond these three primary failure pathways, the proposed PDIA may also experience boundary breakdowns under two other extreme conditions. Firstly, under extremely high excitation frequencies, fluid inertia and dynamic phase lag become dominant. Affected by fluid mass inertia and minor compressibility, the fluid inside the damper's restrictive channels cannot track the rapid piston movement instantaneously, introducing a significant phase lag between velocity and force, which subsequently distorts and widens the force-velocity hysteresis loops. Secondly, sensor failure introduces high-frequency signal distortion and abnormal noise issues. Near the velocity zero-crossing points, differentiated displacement signals suffer from severe distortion, which distorts the input signals and subsequently triggers target current calculation jitter or solver non-convergence.

4.6.2 Limitations of practical application

For practical engineering implementations, analyzing the potential limitations of deploying the model on physical hardware is crucial for future applications. Through analysis, the primary limitations of the proposed PDIA exist as follows:

- a. Sensor noise sensitivity near dead-zones: Near velocity zero-crossings, high-frequency sensor noise can cause numerical jitter when evaluating the inverse division operator. Although the robust division operator with a stabilizing factor ε caps the computational upper bound, sensor filtering and signal conditioning remain critical in practical deployments.
- b. Computational constraints on low-cost hardware: The inverse deduction process relies on adopting an iterative bisection root-finding method within the monotonic latent space. Although the strictly monotonic mapping guarantees rapid convergence within a highly limited number of iterative steps, running this iterative algorithm at a high control frequency may still impose a certain computational burden on automotive-grade microcontrollers with restricted hardware resources.
- c. Actuator inductive delay: Even if the inverse model calculates the target control current I_t instantaneously, the physical current driver (actuator) and electromagnet

coil require finite response time (due to coil inductance) to physically establish the target current. This hardware response delay can lead to a phase lag in force tracking, which must be compensated by external feedforward controllers in practical deployments.

4.6.3 Extended discussion

As demonstrated by the preceding experimental results, the proposed PDIA successfully achieves unified forward-inverse modeling. Analysis indicates that this performance advantage is primarily derived from the balance between rigid physical constraints and data-driven adaptability. The PDIA addresses this trade-off through a decoupled hybrid framework. Instead of blindly imposing rigid constraints across the entire network architecture, strict physical boundaries—such as current monotonicity and macroscopic energy dissipation—are enforced exclusively at the critical control-output interfaces. This configuration ensures a reliable safety net during closed-loop control executions, preventing any non-physical outputs.

Concurrently, the complex dynamic scaling behaviors and hysteretic memory characteristics of the MR fluid are modeled via fully flexible, data-driven deep neural networks within the state observer. This paradigm shift transforms the role of physical laws from the rigid regulation of local parameters to a boundary restriction mechanism for global consistency. By allowing physical laws to govern the macroscopic framework while leveraging the data-driven network to calibrate microscopic details, the architecture successfully bypasses the inherent limitations of both unconstrained black-box models and over-constrained analytical templates, thereby guaranteeing both physical stability and adaptive representational capability.

Within the context of physics-driven modeling, understanding the critical balance of absolute mathematical constraints and managing how dynamic networks adapt near these boundary regimes is of paramount importance [45]. By restricting mathematical templates strictly to the control interfaces while allowing empirical networks to capture transient state transitions, the proposed architecture successfully prevents the model from experiencing non-physical boundary breakdowns under absolute restrictions, establishing a highly adaptive yet physically safe representation of complex nonlinear systems.

5 Conclusion

This paper presents a physics-driven invertible architecture (PDIA) for the integrated forward-inverse modeling of MR dampers. By embedding core physical priors into the

underlying network as hard constraints, this architecture fundamentally addresses the inherent bottlenecks of traditional models regarding physical interpretability, modeling accuracy, and inverse deductibility. Furthermore, governed by global monotonic bijection reconstruction and latent-space alignment joint optimization, the PDIA guarantees the absolute uniqueness of the inverse solution while achieving end-to-end collaborative optimization for both forward and inverse models across all operating conditions.

To systematically validate the comprehensive performance of the PDIA, rigorous multi-dimensional experiments were conducted. Initially, the full-condition evaluation demonstrated the model's capacity to yield unbiased, high-fidelity forward and inverse predictions. Subsequent visual decoupling analysis confirmed that the three core latent variables strictly adhere to predefined design objectives and physical constraints. Correspondingly, the *MAE* for the current and force closed-loop reconstructions remained below 0.018 A and 0.009 kN, respectively, fully verifying the architecture's physical interpretability and intrinsic consistency. When evaluated against baseline models, the inference fidelity of the PDIA significantly outperformed all alternatives in both forward and inverse tasks. More crucially, the proposed architecture exhibited a commanding advantage during zero-shot generalization testing under unseen conditions; the overall *MAE* for the forward damping force prediction was minimized to 0.0170 kN, while the inverse current deduction *MAE* was restricted to merely 0.0211 A. Finally, when subjected to pseudo-random target force commands featuring multi-frequency superpositions and time-varying amplitudes in the hardware tracking experiment, the PDIA accurately tracked the dynamic fluctuations. This conclusively validates its practical feasibility for deployment in real-world semi-active control systems.

In summary, the proposed PDIA mitigates the unconstrained limitations of purely black-box representations in non-linear modeling, offering a robust approach for the bidirectional modeling of magnetorheological dampers. It achieves a balanced coordination of physical consistency, prediction accuracy, and numerical stability. This architecture presents practical value for the unified forward-inverse modeling of smart damping devices, showing potential for semi-active vibration control applications.

Author contributions Author contribution Feng Zhang wrote the original draft, developed the methodology, and contributed to visualization. Minghui Zhu developed the software and conceived the study. Zhiyuan Zou performed formal analysis and conducted investigation. Jiahao Li contributed to visualization and performed data curation. Benyuan Fu supervised the project and acquired funding. Honghui Zhang performed writing-review & editing,

provided resources, and acquired funding. All authors reviewed and approved the final manuscript.

Funding This work was supported by the National Natural Science Foundation of China (No.52375084).

Data availability Data will be made available on reasonable request.

Declarations

Competing interest The authors declare no conflict of interest.

References

- Liu, G., Gao, F., Liao, W.-H.: Design and optimization of a magnetorheological damper based on B-spline curves. *Mech. Syst. Signal Process.* **178**, 109279 (2022)
- Hu, G., Ying, S., Qi, H., Yu, L., Li, G.: Design, analysis and optimization of a hybrid fluid flow magnetorheological damper based on multiphysics coupling model. *Mech. Syst. Signal Process.* **205**, 110877 (2023)
- Nguyen, S.D., Choi, S.-B., Nguyen, Q.H.: A new fuzzy-disturbance observer-enhanced sliding controller for vibration control of a train-car suspension with magneto-rheological dampers. *Mech. Syst. Signal Process.* **105**, 447–466 (2018)
- Li, Z., Yao, J., Shi, W., Fang, M.: A novel method of vehicle height control utilizing semi-active actuator. *Mech. Syst. Signal Process.* **223**, 111818 (2025)
- Yu, S., Guo, J., Xu, X., Zhang, S., Lin, B.: A cascaded control strategy for magneto-rheological dampers based on Hammerstein model. *Mech. Syst. Signal Process.* **222**, 111748 (2025)
- Ma, B., Huang, H., Huang, J., Lin, X., Huang, Q., Chen, S.: A forward-inverse dynamic model for the hydraulic damping(-magnetorheological) actuator based on cyclic stress-strain. *Smart Mater. Struct.* **33**, 025001 (2024)
- Majdoub, K.E., Giri, F., Chaoui, F.-Z.: Adaptive backstepping control design for semi-active suspension of half-vehicle with magnetorheological damper. *IEEE/CAA J. Autom. Sin.* **8**, 582–596 (2021)
- Chen, K.-Y.: A new state observer-based vibration control for a suspension system with magnetorheological damper. *Veh. Syst. Dyn.* **60**, 3127–3150 (2022). <https://doi.org/10.1080/00423114.2021.1942508>
- Xing, X., Chen, Z., Yu, D., Feng, Z., Liu, Y.: An invertible hysteresis model for magnetorheological damper with improved adaption capability in frequency and amplitude. *Smart Mater. Struct.* **33**, 055006 (2024)
- Pei, P., Peng, Y., Qiu, C.: An improved semi-active structural control combining optimized fuzzy controller with inverse modeling technique of MR damper. *Struct Multidisc Optim* **65**, 272 (2022)
- Chen, G., Xiong, X., Lou, Y., Li, Z.: Modeling and observation of rate-dependent hysteresis and creep phenomena in magnetorheological clutch. *IEEE/ASME Trans. Mechatron.* **27**, 2053–2061 (2022)
- Zhao, J., Deng, L., Ning, D., Zhang, H., Sun, S., Li, W., Du, H.: Modeling and torque control against rate-dependent hysteresis of a magnetorheological fluid dual clutch in an electric vehicle transmission system. *IEEE/ASME Trans. Mechatron.* **30**, 27–37 (2025)
- Guo, Q., Yang, X., Li, K., Li, D.: Parameters identification of magnetorheological damper based on particle swarm optimization algorithm. *Eng. Appl. Artif. Intell.* **143**, 110016 (2025)
- Hong, S.R., Wereley, N.M., Choi, Y.T., Choi, S.B.: Analytical and experimental validation of a nondimensional Bingham model for mixed-mode magnetorheological dampers. *J. Sound Vib.* **312**, 399–417 (2008)
- Dimock, G.A., Yoo, J.-H., Wereley, N.M.: Quasi-steady Bingham biplastic analysis of electrorheological and magnetorheological dampers. *J. Intell. Mater. Syst. Struct.* **13**, 549–559 (2002)
- Ismail, M., Ikhrouane, F., Rodellar, J.: The hysteresis Bouc-Wen model, a survey. *Arch. Comput. Methods Eng.* **16**, 161–188 (2009)
- Spencer, B.F., Dyke, S.J., Sain, M.K., Carlson, J.D.: Phenomenological model for magnetorheological dampers. *J. Eng. Mech.* **123**, 230–238 (1997)
- Awrejcewicz, J., Dzyubak, L., Lamarque, C.-H.: Modelling of hysteresis using Masing–Bouc-Wen’s framework and search of conditions for the chaotic responses. *Commun. Nonlinear Sci. Numer. Simul.* **13**, 939–958 (2008)
- Zhang, S., Shi, W., Chen, Z.: Modeling and Parameter Identification of MR Damper considering Excitation Characteristics and Current. *Shock. Vib.* **2021**, 6691650 (2021)
- Cheng, M., Chen, Z., Liu, W., Jiao, Y.: A novel parametric model for magnetorheological dampers considering excitation characteristics. *Smart Mater. Struct.* **29**, 045002 (2020)
- Choi, S.-B., Lee, S.-K., Park, Y.-P.: A hysteresis model for the field-dependent damping force of a magnetorheological damper. *J. Sound Vib.* **245**, 375–383 (2001)
- Bai, X.-X., Cai, F.-L., Chen, P.: Resistor-capacitor (RC) operator-based hysteresis model for magnetorheological (MR) dampers. *Mech. Syst. Signal Process.* **117**, 157–169 (2019)
- Feng, Z., Yu, D., Chen, Z., Xing, X., Yan, H.: A new hysteresis model with resistor-capacitor operator for magnetorheological dampers. *J. Vib. Eng. Technol.* **11**, 3229–3244 (2023)
- Bai, X.-X., Li, C.-X.: Precise real-time hysteretic force tracking of magnetorheological damper. *Smart Mater. Struct.* **29**, 104002 (2020)
- Jiang, R., Rui, X., Wei, M., Yang, F., Zhu, H., Gu, L.: A phenomenological model of magnetorheological damper considering fluid deficiency. *J. Sound Vib.* **562**, 117851 (2023)
- Zhang, G., Luo, J., Sun, M., Yu, Y., Chen, J., Wang, J., Ouyang, Q., Qiu, Y., Chen, G., Liu, Q., Chen, B., Shen, T., Zhang, Z.: A novel reliable parametric model for predicting the nonlinear hysteresis phenomenon of composite magnetorheological fluid. *Smart Mater. Struct.* **34**, 035060 (2025)
- Miah, M.S., Chatzi, E.N., Dertimanis, V.K., Weber, F.: Nonlinear modeling of a rotational MR damper via an enhanced Bouc–Wen model. *Smart Mater. Struct.* **24**, 105020 (2015)
- Chen, G., Lou, Y.: Recurrent-neural-network-based rate-dependent hysteresis modeling and feedforward torque control of the magnetorheological clutch. *IEEE/ASME Trans. Mechatron.* **27**, 2875–2886 (2022)
- Lu, Y., Li, H., Lai, J., Wang, J., Shi, K., Lu, T., Song, A.: Multidimensional hysteresis modeling and analysis of rate-dependence and creep phenomena in magnetorheological actuator. *IEEE/ASME Trans. Mechatron.* **30**, 7322–7333 (2025)
- Bai, X.-X., Chen, P., Qian, L.-J.: Principle and validation of modified hysteretic models for magnetorheological dampers. *Smart Mater. Struct.* **24**, 085014 (2015)
- Saafi, M.S.R.M., Talib, M.H.A., Zain, M.ZMd.: The Application of Machine Learning and Optimisation Algorithm for Magnetorheological Damper Dynamics Behaviour: A Review. *J. Vib. Eng. Technol.* **13**, 231 (2025)
- Jiang, M., Rui, X., Yang, F., Wang, P., Li, H., Wang, W.: Non-parametric modeling of magnetorheological damper based on MEA-BP neural network. *J. Vib. Control* **32**, 1838–1851 (2026)
- Lv, H., Sun, Q., Ma, W.: Data-driven prediction-control system modeling for magnetorheological damping force. *J. Intell. Mater. Syst. Struct.* **34**, 155–167 (2023)

34. Delijani, Y.M., Cheng, S., Gherib, F.: Sequential neural network model for the identification of magnetorheological damper parameters. *Smart Mater. Struct.* **33**, 015002 (2024)
35. Wei, S., Wang, J., Ou, J.: Method for improving the neural network model of the magnetorheological damper. *Mech. Syst. Signal Process.* **149**, 107316 (2021)
36. Wang, D.H., Liao, W.H.: Modeling and control of magnetorheological fluid dampers using neural networks. *Smart Mater. Struct.* **14**, 111 (2004)
37. Li, J., Luo, J., Zhang, F., Zhou, W., Wei, X., Liao, C., Shou, M.: Modeling of magnetorheological dampers based on a dual-flow neural network with efficient channel attention. *Smart Mater. Struct.* **32**, 105006 (2023)
38. Chen, X., Yao, Y., Su, H., Qiao, J.: Modeling and TF-BPNN inverse modeling of a magnetorheological damper. *J. Vib. Control* **0**, 1–19 (2026)
39. Lei, Y., Yang, B., Jiang, X., Jia, F., Li, N., Nandi, A.K.: Applications of machine learning to machine fault diagnosis: a review and roadmap. *Mech. Syst. Signal Process.* **138**, 106587 (2020)
40. Guan, T., Gao, D., Ma, J., Wu, J., Wang, S., Zhang, C., Wang, L., Tang, J.: Interpretable compound fault diagnosis of hydraulic gear pumps via fusion of physical priors and probabilistic joint reasoning. *Mech. Syst. Signal Process.* **243**, 113698 (2026)
41. Dong, C., Tao, J., Sun, H., Wei, Q., Tan, H., Liu, C.: Innovative fault diagnosis for axial piston pumps: a physics-informed neural network framework predicting pump flow ripple. *Mech. Syst. Signal Process.* **225**, 112274 (2025)
42. Fang, X., Song, Q., Wang, X., Li, Z., Ma, H., Liu, Z.: An intelligent tool wear monitoring model based on knowledge-data-driven physical-informed neural network for digital twin milling. *Mech. Syst. Signal Process.* **232**, 112736 (2025)
43. Bai, J., Lin, Z., Wang, Y., Wen, J., Liu, Y., Rabczuk, T., Gu, Y., Feng, X.-Q.: Energy-based physics-informed neural network for frictionless contact problems under large deformation. *Comput. Methods Appl. Mech. Eng.* **437**, 117787 (2025)
44. Chen, Z., Lai, S.-K., Yang, Z.: AT-PINN: advanced time-marching physics-informed neural network for structural vibration analysis. *Thin-Walled Struct.* **196**, 111423 (2024)
45. Chen, G.: From equation to expression: positioning narrative as the epistemic pilot beyond mathematics and physics. *Tone Uprising J. Narrat. Sover.* (2025). <https://doi.org/10.5281/zenodo.15765807>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.